LYRA CAPITAL

User guide

What each screen does, in which order to use them, and what the numbers mean.

How to read it. Each chapter is one area of the application. Read it from the start, or go straight to the chapter for the screen in front of you.

To change Lyra, rather than use it, there is another document: the *Guida dello sviluppatore* (developer guide, in Italian), in the same folder.

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Before you start

Lyra is for building investment strategies, testing them on past data and — when they hold up — running them on a real account. The three happen in that order, and this guide follows the same order.

In the text, **This** is a menu item or a tab, This a button to press, *this* a field to fill in, and Settings → Data means: open **Settings**, then the **Data** tab.

The boxes

Note

A clarification. Skip it and the operation still works.

In practice

The operation, step by step.

Warning

Something can be lost here: data, money or time. Worth reading first.

A word about the numbers

No result shown by the application is a forecast. A backtest says how a strategy *would have* behaved on past data, with estimated costs; it does not say how it will behave. The chapter on **Streams** explains which numbers to look at to tell a strategy that worked from one that got lucky.

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1

Getting started

The numbers in the screenshots are made up

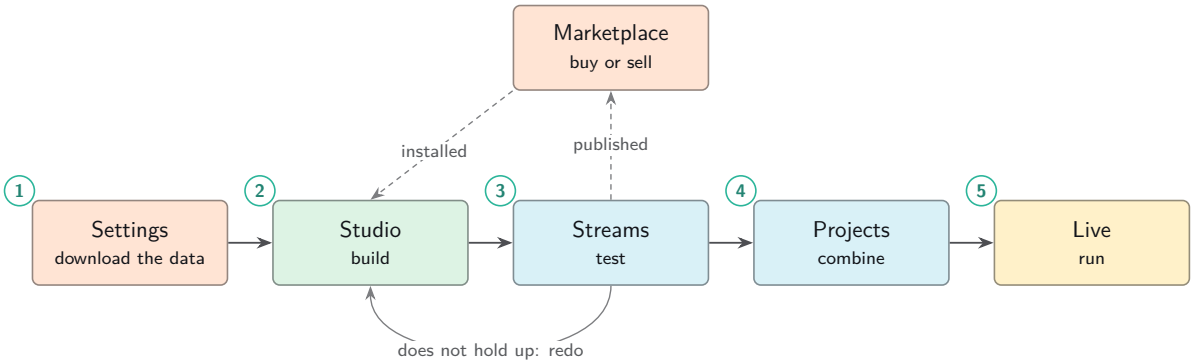
Every image in this guide was taken on an installation with *sample data*: series generated by a computer, not real prices and not the results of real strategies. They are there to show where things are. No Sharpe ratio, return or drawdown on these pages means anything.

1.1 What Lyra does

Three things, in this order:

1. **Build** an investment strategy — by writing it, by drawing it with blocks, or by describing it in words.
2. **Test** it on past data, with costs, and see how it would have behaved.
3. **Run** it on a real account, when it holds up.

Strategies can also be combined with one another, and bought and sold between users on the marketplace of the **Community**. This guide follows the same order: one chapter per area of the application.



Only the Marketplace, in the Community, needs an account. Everything else runs on your own computer.

Figure 1.1. The areas of the application, in the order you use them. The arrow that goes back is the most frequent part of the work.

1.2 The menu

The bar on the left is always there. It widens when you hover over it with the mouse and shows the names of the items.

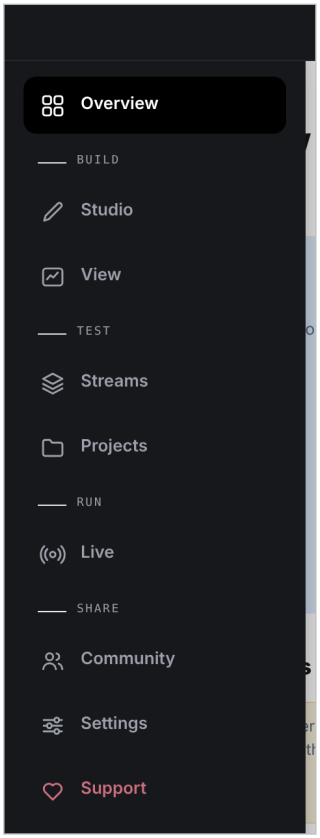


Figure 1.2. The menu, widened. Below **Overview** the items are in four groups — **Build**, **Test**, **Run**, **Share**. **Settings** sits at the bottom.

Item	Group	What it is for
Overview	—	The home screen: how everything is going, on one page.
Studio	Build	Where strategies are built.
View	Build	The charts, with what the strategy did drawn on top.
Streams	Test	The results: one card per strategy.
Projects	Test	Combinations of several strategies.
Live	Run	The link to a real account.
Community	Share	The strategy marketplace, messages, feedback and your own account.
Settings	—	Data, parameters, market regimes.

Top right

The magnifying glass searches the whole application; the bell shows finished jobs and messages; the sun switches between light and dark; the last item is the way into your account.

1.3 Do you need an account?

To build, test and run strategies, **no**: the application runs on your own computer and asks for nothing.

An account is needed only for the areas that go through the network: the **Community** —

marketplace, messages, feedback, profile — and the subscription. If you open one of them without being signed in, the application says so and offers to sign you in.

1.4 The order in which you work

From nothing to the first result

1. Settings → Data: download the data. Without data, every page is empty — it is not an error, there is simply nothing to show.
2. Streams → Run backtest: choose a strategy and press **Run backtest**.
3. Wait. The job runs in the background and the bell tells you when it is done; meanwhile you can use the rest of the application.
4. **Streams**: the result has appeared as a card.
5. **Projects**: combine two or more results.
6. **Live**: connect a broker, when you are ready.

2

Overview

It is the page that opens first. It answers a single question: how is everything going, right now.

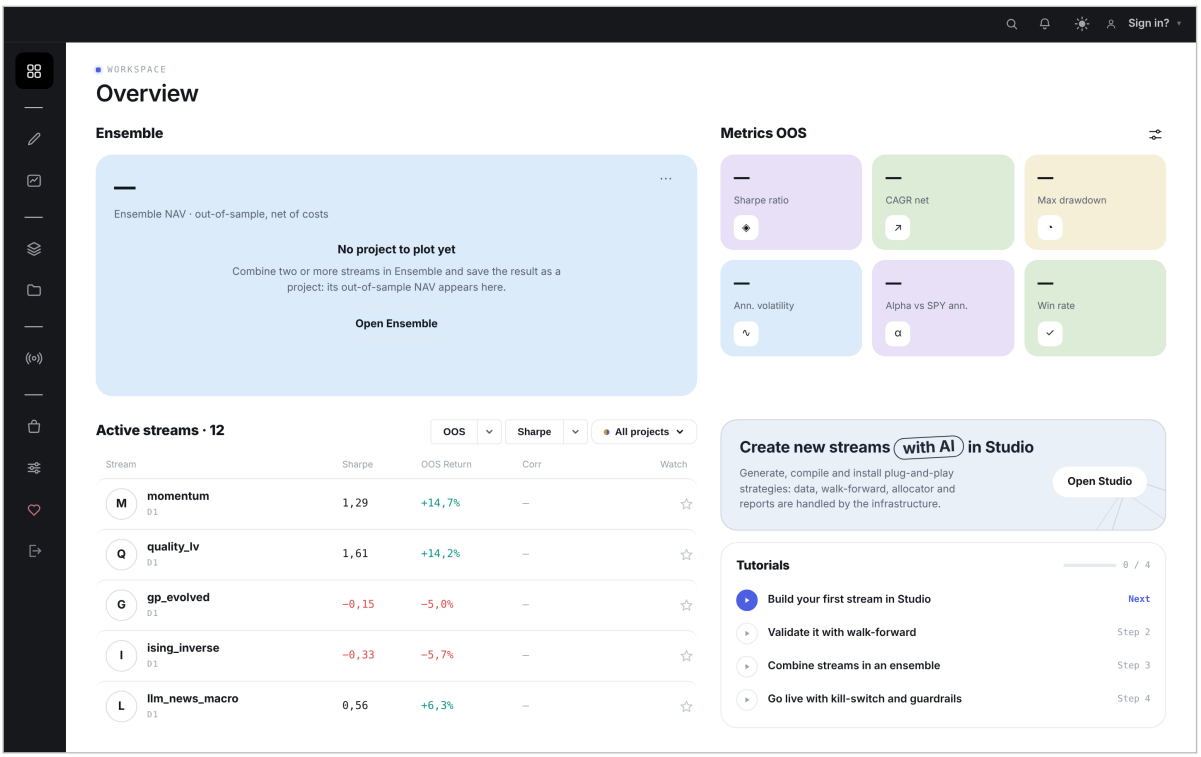


Figure 2.1. The home screen. Four panels: the ensemble chart, the metrics, the list of active streams, and the shortcut to Studio.

2.1 The chart at the top left

It shows the performance of the chosen **Project**: the combination of several strategies, net of estimated costs, on the data the strategy had *not* seen while it was being built.

Until a project has been saved, an invitation to create one takes its place. It is not an error: there is simply nothing to draw yet.

2.2 Metrics OOS

Six numbers, and they are meant to be read together. **OOS** stands for *out of sample*: computed on data the strategy did not know.

Metric	What it tells you
Sharpe ratio	How much it earns relative to how much it swings. It is the number most used to compare two strategies, and the easiest to make look good by looking at the wrong data.
CAGR net	The compound annual return, already net of estimated costs.
Max drawdown	The largest loss from a previous peak. It is the question 'how much would I have had to endure before recovering'.
Ann. volatility	How much it swings in a year.
Alpha vs SPY ann.	How much it earns *beyond* the market, once the part explained by the market itself is removed. A strategy with a good return and an alpha close to zero is buying the market under other names.
Win rate	The percentage of days in profit. On its own it says little: you can win nine times out of ten and still lose.

Which numbers to look at first

Max drawdown and **Alpha vs SPY**. The first because it is the figure that decides whether a strategy can be lived with: an excellent Sharpe with a 45% drawdown is a strategy nobody holds to the end. The second because it tells a strategy that does something apart from one that follows the market with longer strides.

A backtest is not a forecast

It says how a strategy *would have* behaved on past data, with estimated costs. It does not say how it will behave. Chapter 4 explains which signs tell a strategy that worked from one that got lucky.

2.3 Active streams

The list of all available results, sorted. The columns:

Sharpe
As above.

OOS Return
The out-of-sample return. Green above zero, red below.

Corr
The correlation with the other streams. It is the column that matters most when combining: two highly correlated strategies, put together, diversify nothing — they make the same bet twice.

Watch
The star adds the stream to your favourites.

The three menus above the list choose the period (**OOS** or all of it), the sort order and the reference project.

2.4 Tutorials

At the bottom right, four steps that tick themselves off as you do them: build the first stream, validate it, combine it, put it into production. It is the path this guide follows in the next chapters.

3

Studio

Studio is where things are built. You get there from **Studio** in the menu, or from the **Open Studio** button on the home screen.

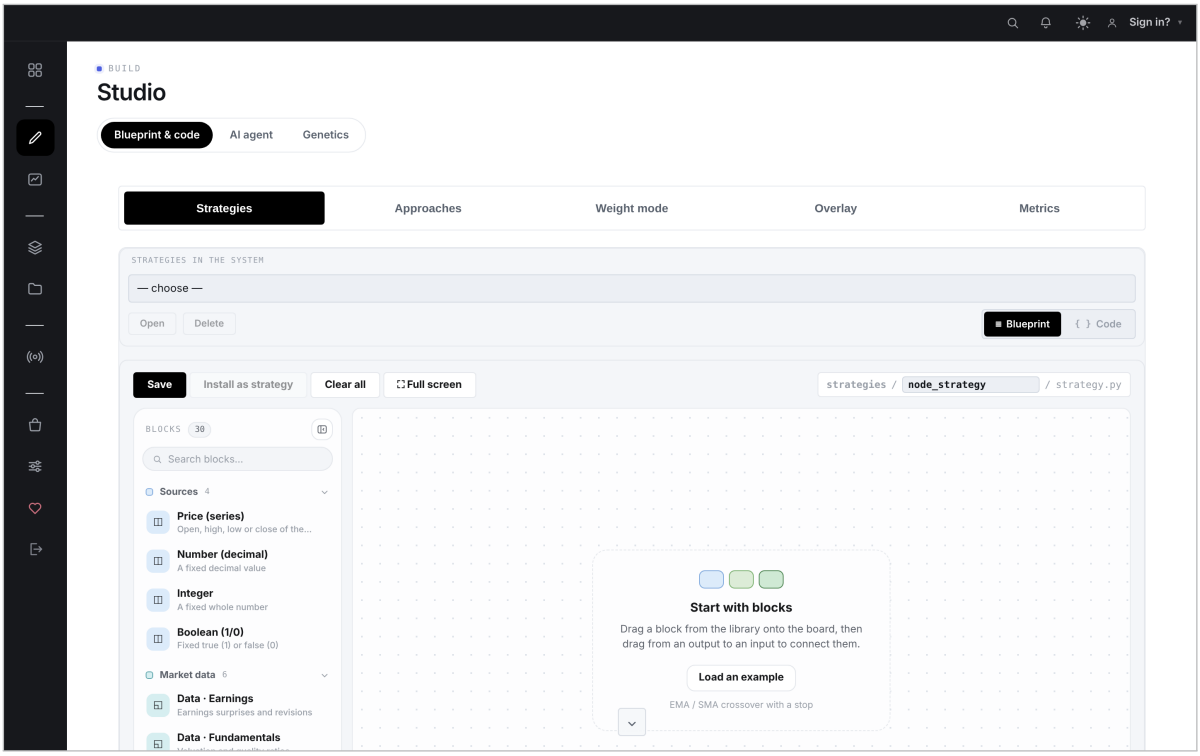


Figure 3.1. Studio. At the top the three ways of working, below them the five families of artifacts, in the middle the drawing board.

3.1 The five families

The row of tabs below the title says *what* you are building. They are five different things, and it is worth knowing which one you need before you start.

Tab	What it is
Strategies	A strategy: it decides what to buy and sell, and when. It is the thing you build most often.
Approaches	An approach: it skips the whole engine and produces the series of gains and losses directly. It is for people who already have a calculation of their own and just want to bring it into the system.
Weight mode	A rule for sharing capital among several strategies. Fourteen come already installed.
Overlay	A brake: it reduces exposure when the market behaves in a certain way. It does not earn, it protects.
Metrics	A measure of your own, a number or a chart, shown on the page of every stream next to the default ones. It can read all the data loaded in the system (Section 3.3).

3.2 The three ways to build

At the top right you choose how to work. The result is the same — a code file installed in the system — and only the road changes.

3.2.1 Blueprint: with blocks

You drag blocks from the library on the left onto the board and connect outputs to inputs. The library is divided into categories: prices, numbers, external data such as earnings and fundamentals.

> **The quickest way to see how it works**
>
> Press **Load an example**: an example appears, already wired (the crossing of two moving averages with a stop). Take it apart and put it back together: in ten minutes you understand the mechanism without writing a line.

3.2.2 Code: by writing

The **{ } Code** button switches to the text editor. It is the same artifact, seen as Python code. You can switch between the two views.

3.2.3 AI agent: in words

The **AI agent** tab opens a conversation: you describe the strategy in words and the model writes it. It needs a model installed locally; without one, the tab says so and the other two ways keep working.

> **Read what the assistant writes**
>
> The generated code is installed in the system and runs on your data — and one day, perhaps, on your account. The assistant does not know whether a strategy makes financial sense: it knows how to write code that does what it was asked. The **Code** tab is there for this too.

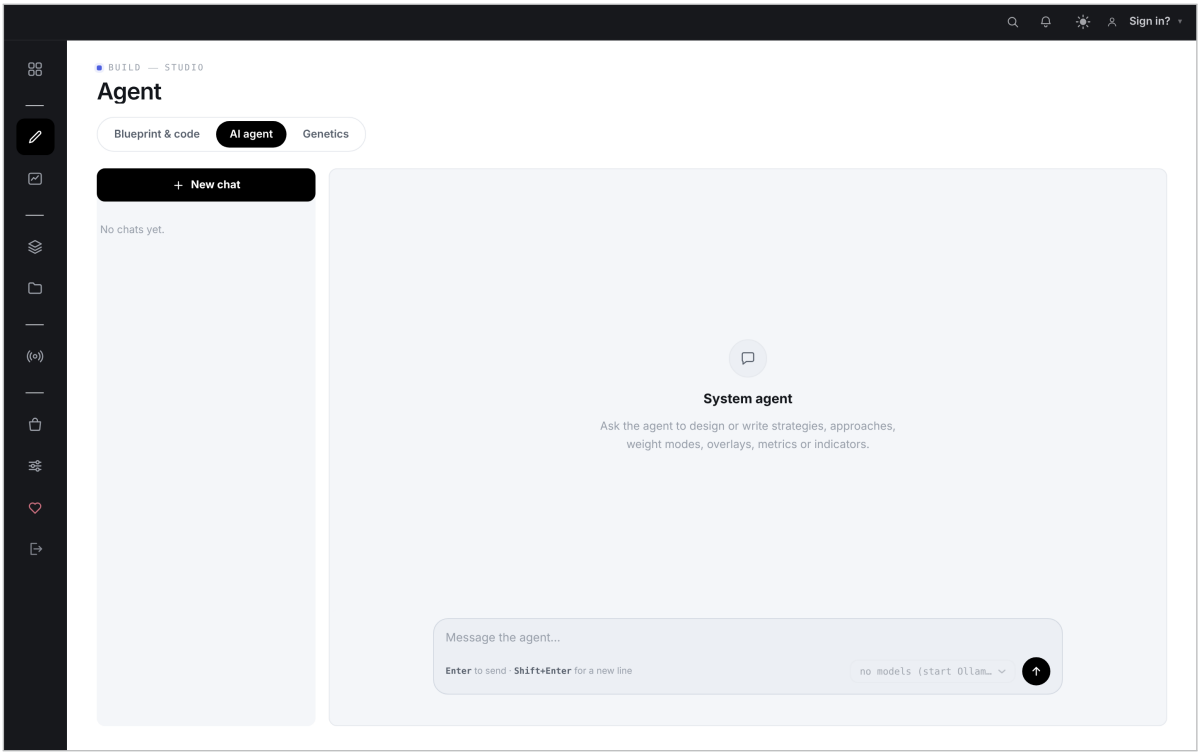


Figure 3.2. The assistant. You describe what you want, and it writes the artifact and installs it.

3.3 Metrics

A metric is a few lines of Python on the results of a strategy. Two fields above the editor say what it produces: **Output** — **Number** or **Chart** — and, for a chart, **Chart type**. **Figure** is the type for a chart with several series, points and an axis of numbers instead of dates.

A number: beta against the S&P 500

```
r_m = data.assets.returns("SPY", timeframe="D1")
both = returns.notna() & r_m.notna()
result(returns[both].cov(r_m[both]) / r_m[both].var(), "Beta vs SPY", fmt="ratio")
```

`returns` is the strategy's own series; `data` is everything loaded in the system: prices, earnings, fundamentals, rates, macro series, news, and the categories you created in Settings → Data. Where the data comes in several timeframes, the metric reads the one of the strategy, the lowest if it uses more than one; `timeframe="D1"` picks another. The full list, with a CAPM chart as an example, is in the **Template** of the editor and in the guide `metrics/METRIC_TEMPLATE.md`.

Save checks the code and saves it in one go: if something is wrong, the reason appears under the editor and nothing is saved. **Validate** checks without saving. If the code draws charts while **Output** says **Number**, or the other way round, the output is corrected by itself, and **Chart type** follows what the code draws.

Saving a metric whose data is missing

If the metric reads something that is not in the system yet, Studio says so and saves it anyway: it is not an error in the code. On the stream page it shows **Data not available** until the data arrives.

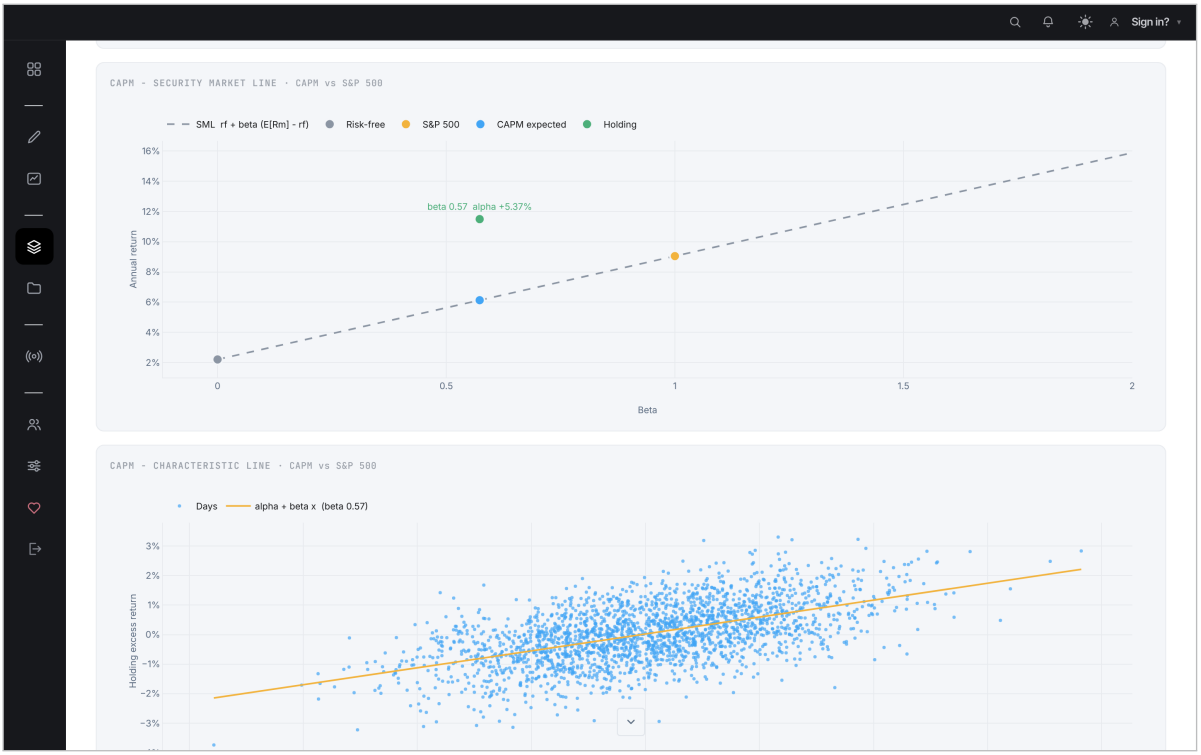


Figure 3.3. A **Figure** metric on the page of a stream: the CAPM against the S&P 500. The dashed line is what the market pays for each unit of beta; the strategy, above it, earned more than its beta was worth.

3.4 Genetics

The **Genetics** tab opens an area of its own: instead of writing one strategy, you let many of them be born and keep those that survive. You choose the universe of instruments and let it run.

> **It takes a long time**
>
> It is the slowest job in the application. It starts as a background process: you can close the page, and the bell tells you when it is finished.

3.5 What happens when you press ‘Install’

The artifact is written to a folder of the system, and from then on it appears in the lists: a strategy among those you can run, a weighting mode among those you can choose in a project, an overlay among those you can switch on.

Before writing, the system checks the code: it must not read or write files, must not open connections and must not run commands. If it contains any of these, the installation is refused and the reason is shown.

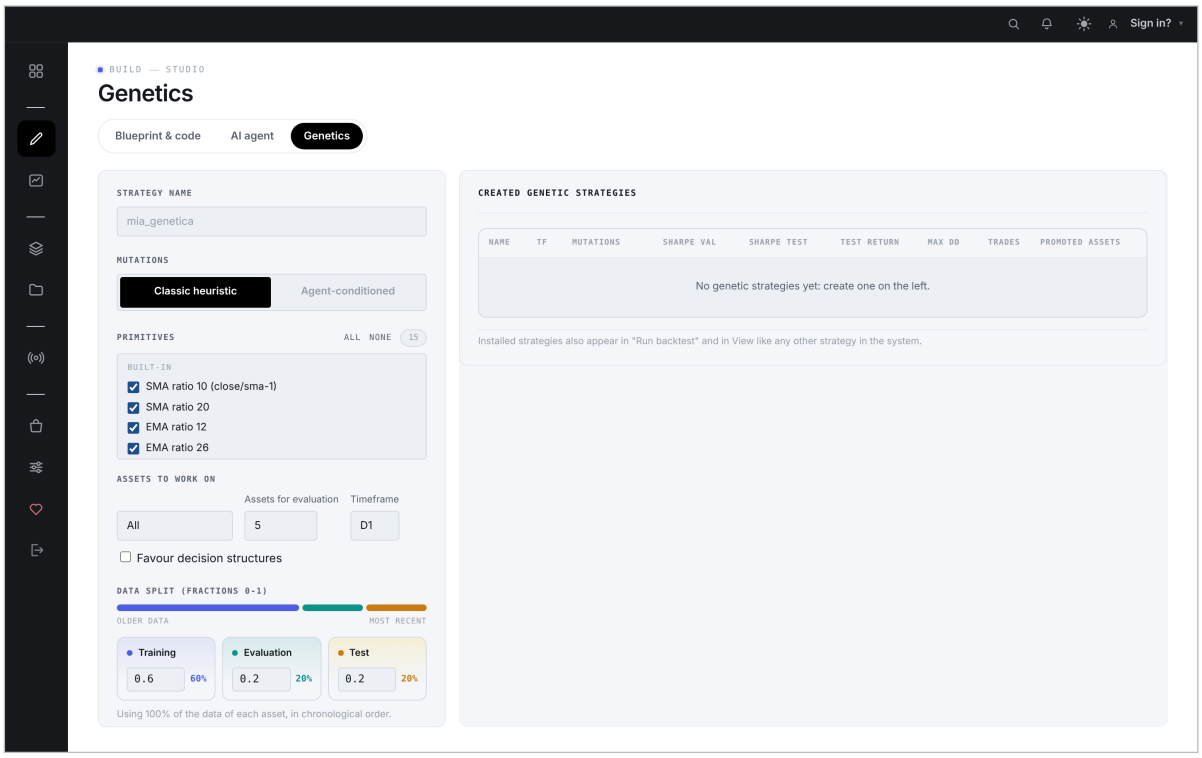


Figure 3.4. The genetics section. You declare what the strategies should evolve on, and the rest is waiting.

4

Streams

A **stream** is the result of a strategy: the series of its gains and losses over time. The **Streams** page shows them all, one card each.

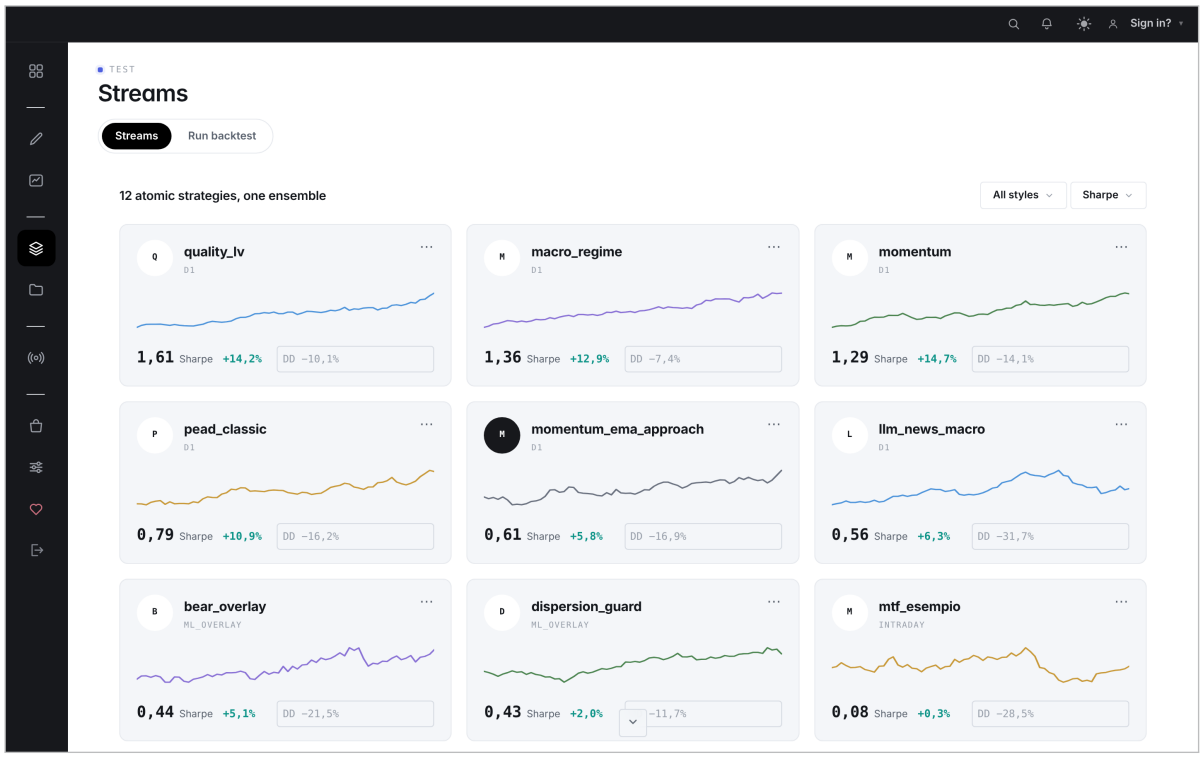


Figure 4.1. The list of results. Each card carries the performance, the Sharpe ratio, the return and the maximum loss.

4.1 Running a backtest

The **Run backtest** tab, next to **Streams**, lists everything that can be run: the installed strategies and approaches, each with its own description and a button.

Running it, and then

1. Press the button of the strategy you want.
2. The page shows the log live: you see what it is doing, line by line.
3. You can close the page. The job carries on, and the bell at the top right tells you when it is done.
4. The result appears in **Streams**.

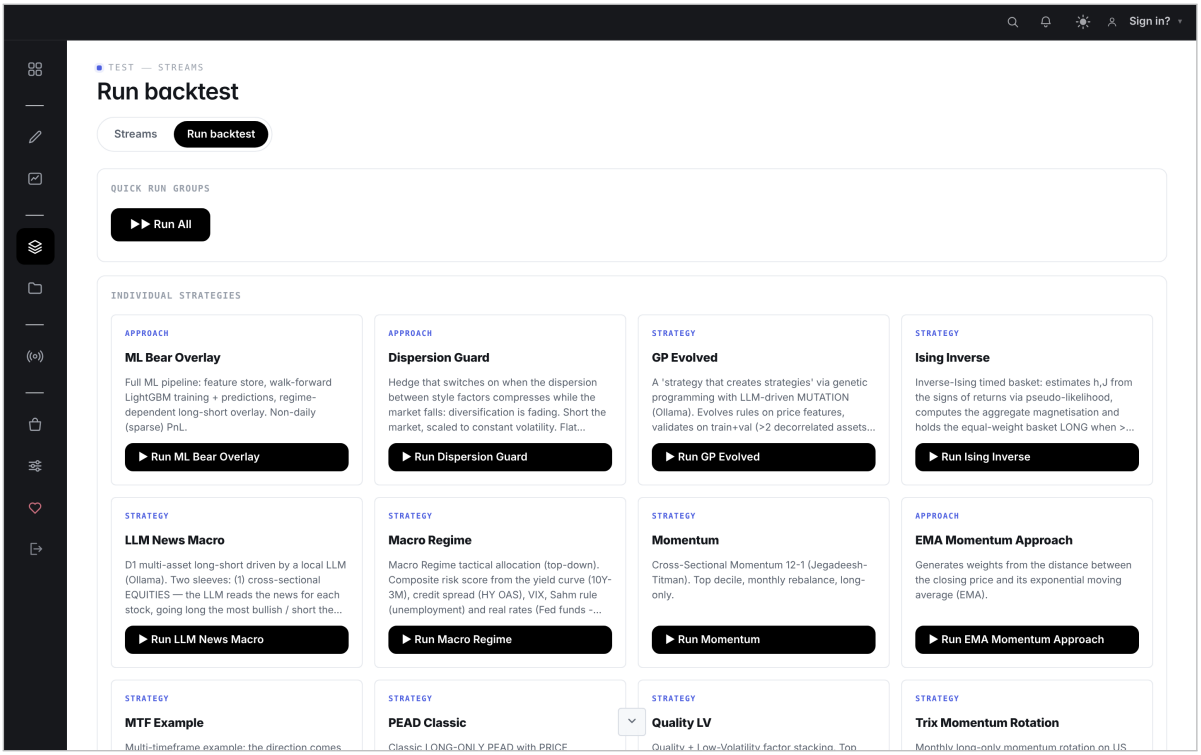


Figure 4.2. What can be run. **Run All** queues them all, one after the other.

One job at a time

Backtests run in a queue, not in parallel, even if you press several buttons. It is not a limit to work around: two calculations at the same time would fight over the same files and produce mixed-up results.

4.2 The page of a result

Clicking on a stream opens its page.

4.2.1 Robustness: the panel to read first

It is the most important part of the page, and it is the one that answers the question 'did this strategy work, or was it lucky?'. The score on the right sums up the five numbers in one.

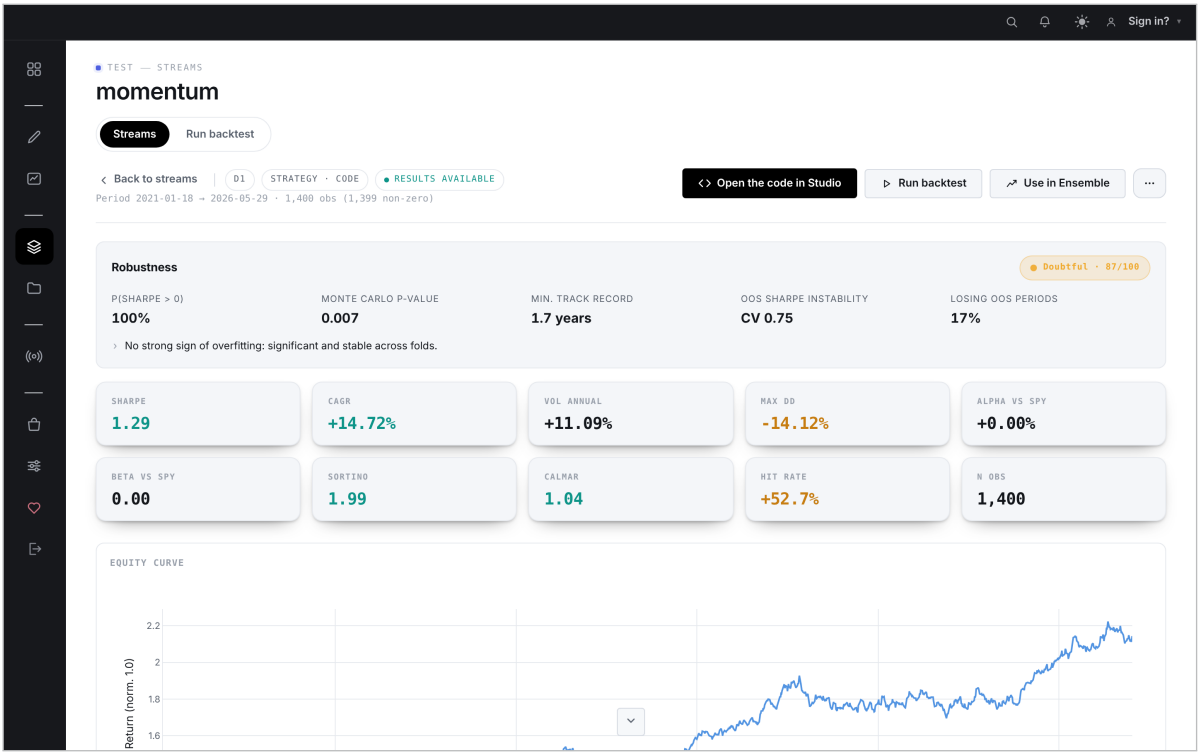


Figure 4.3. The page of a stream. At the top the robustness verdict, below it the metrics, then the curve.

Indicator	What it tells you
P(Sharpe > 0)	The probability that the result is not noise. Below 95% the strategy has proved nothing.
Monte Carlo p-value	The same question asked another way: shuffling the data, how often would a lucky case have done just as well. The smaller, the better.
Min. track record	How long this strategy would need to run for its result to be credible. A value longer than the period tested means: there is not enough history yet.
OOS Sharpe instability	How much the Sharpe ratio changes from one period to the next. High means the strategy worked in some periods and not in others.
Losing OOS periods	The percentage of periods closed at a loss.

Why a 'Doubtful' verdict is not an error

It means the data are not enough to tell this strategy apart from chance. Not that it is wrong: that it has not been proved. It is the honest answer over a short period, and the way to change it is to have more history — not to change the parameters until the verdict improves, which is the most common way of fooling yourself.

4.2.2 The metrics

Ten numbers, unless you choose otherwise (see below). Besides those already seen on the home screen (Chapter 2):

Beta vs SPY
How closely the strategy follows the market. Close to zero: it does something of its own. Close to one: it moves like the index.

Sortino
Like the Sharpe ratio, but it counts only the downward swings. Rising fast is not a flaw.

Calmar
Annual return divided by maximum loss. It answers 'how much does it earn for each point of pain'.

Hit rate
The percentage of periods in profit.

N obs
How many observations. Few observations make everything else less reliable.

4.2.3 Choosing the metrics

Every metric on the page can be hidden: the numbers, the robustness panel, the charts, the tables, and the metrics you wrote yourself in Studio. Move the pointer over one and a small eye appears in its corner: pressing it hides that metric, and the message that follows has an **Undo**. To see the whole list, press **Choose metrics**, at the top of the metrics.

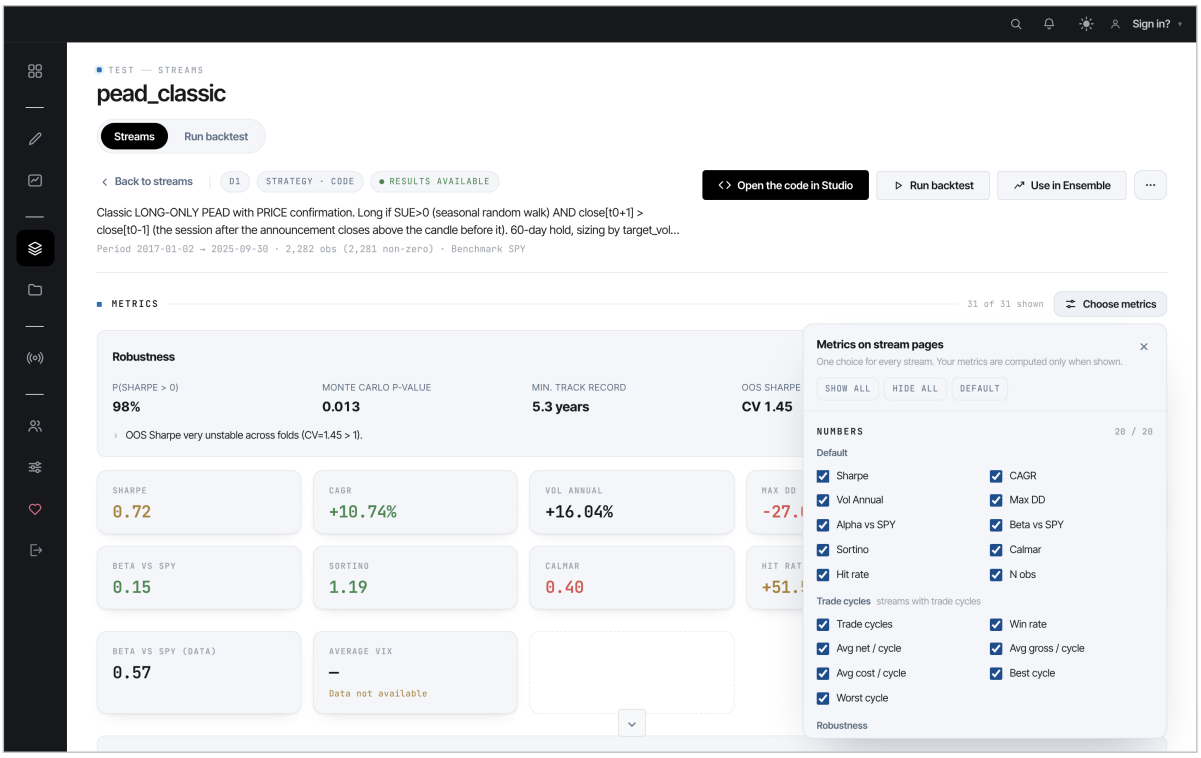


Figure 4.4. **Choose metrics**. Numbers, charts and tables, the default ones and yours, each with its tick. Next to the button, how many are shown.

- The choice holds for *every* stream: you make it once.
- **Show all**, **Hide all** and **Default** change everything at once. **Default** is how the page looks before any choice: the default metrics and your active ones.
- A metric you create after making your choice appears by itself; you hide it like the others

if you do not want it.

- Your metrics are computed only when they are shown, on the results of the stream and with its benchmark.

One of your metrics can show a word instead of a value. **Data not available**: it reads data that is not in the system yet — load it in Settings → Data and the metric works. **Error in the code**: the metric has to be fixed in Studio, and the word takes you there. **No results yet**: the stream has never been run.

> **Adding a metric from here**
>
> Move the pointer after the numbers, or at the very bottom of the page, below the charts and the tables: New number metric and New chart metric appear. They open the **Metrics** section of Studio with the output already chosen, a template and a free name; save, and the metric is on the stream page (Section 3.3) — your numbers after the default ones, your charts at the bottom, just above New chart metric. The pencil on one of your metrics opens it in Studio, and Studio has a link back to the stream.

4.2.4 The three buttons at the top

- Open the code in Studio — opens the artifact in the editor.
- Run backtest — runs the calculation again.
- Use in Ensemble — takes this stream into the project builder (Chapter 5).

4.3 View: seeing what it did

The **View** item in the menu shows prices with the strategy's trades on top: where it went in, where it came out, and the indicators it was watching.
It is for something the numbers do not say: *whether the strategy is doing what you thought.* A curve that rises nicely can hide twenty sensible trades and a single huge lucky one.

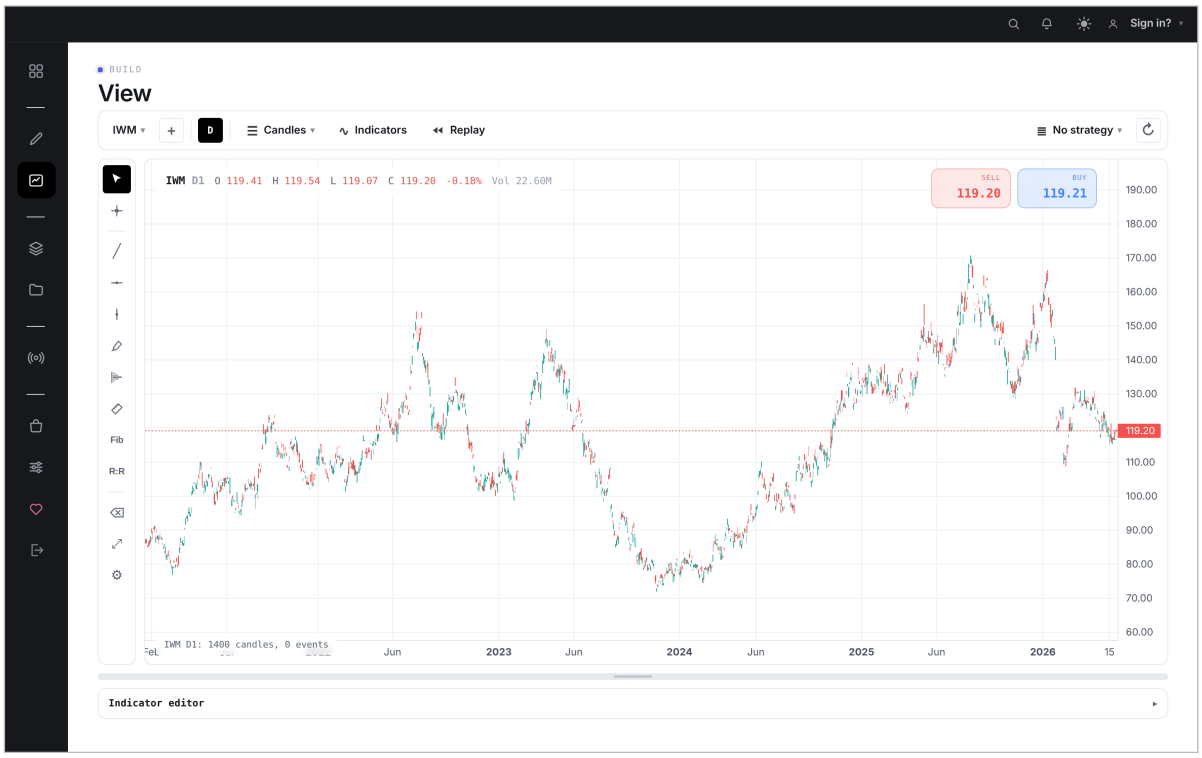


Figure 4.5. The chart view. You choose the strategy and the instrument, and overlay the indicators.

5

Projects

A **project** is a combination of several strategies, with the rule that decides how much each one weighs and any brakes. It is what you put into production: running a single strategy is rarely a good idea.

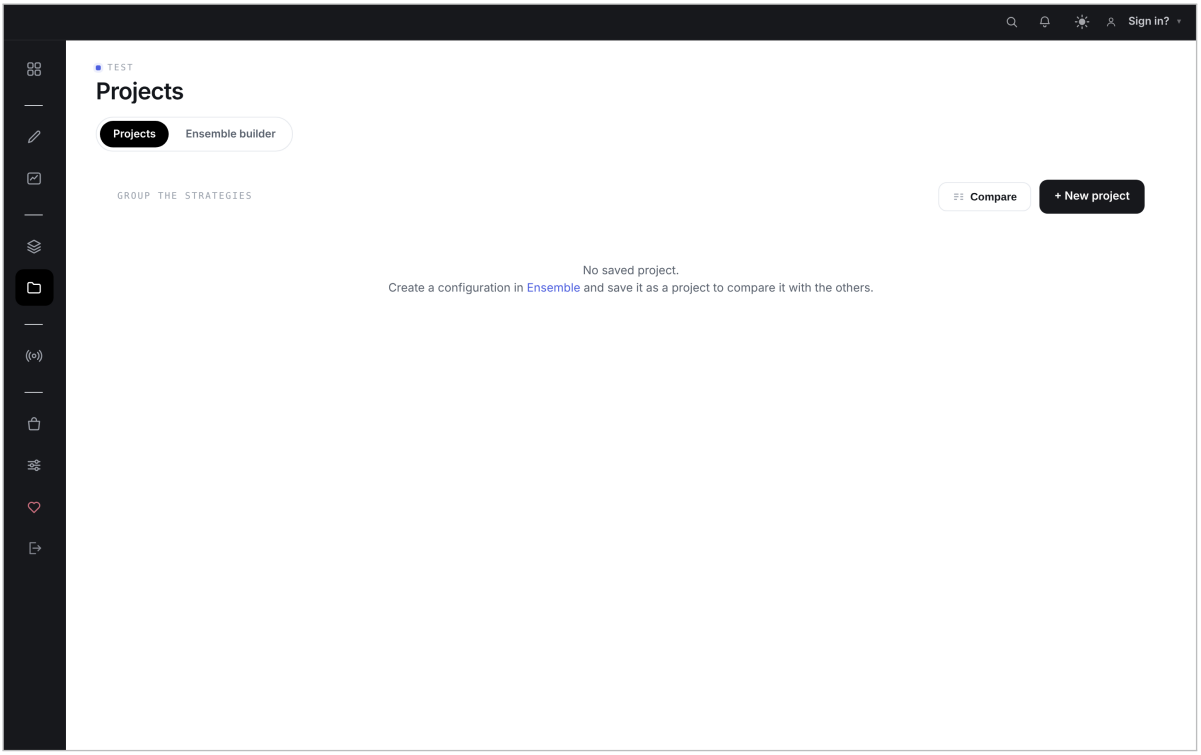


Figure 5.1. The list of saved projects.

5.1 The builder, in four steps

It opens with **Ensemble builder**, or with Use in Ensemble from the page of a stream. The four steps at the top always say where you are.

5.1.1 Step 1 — Strategies

You choose the streams to combine. You need at least two: with one there is nothing to combine, and the Next button stays off until two are chosen.

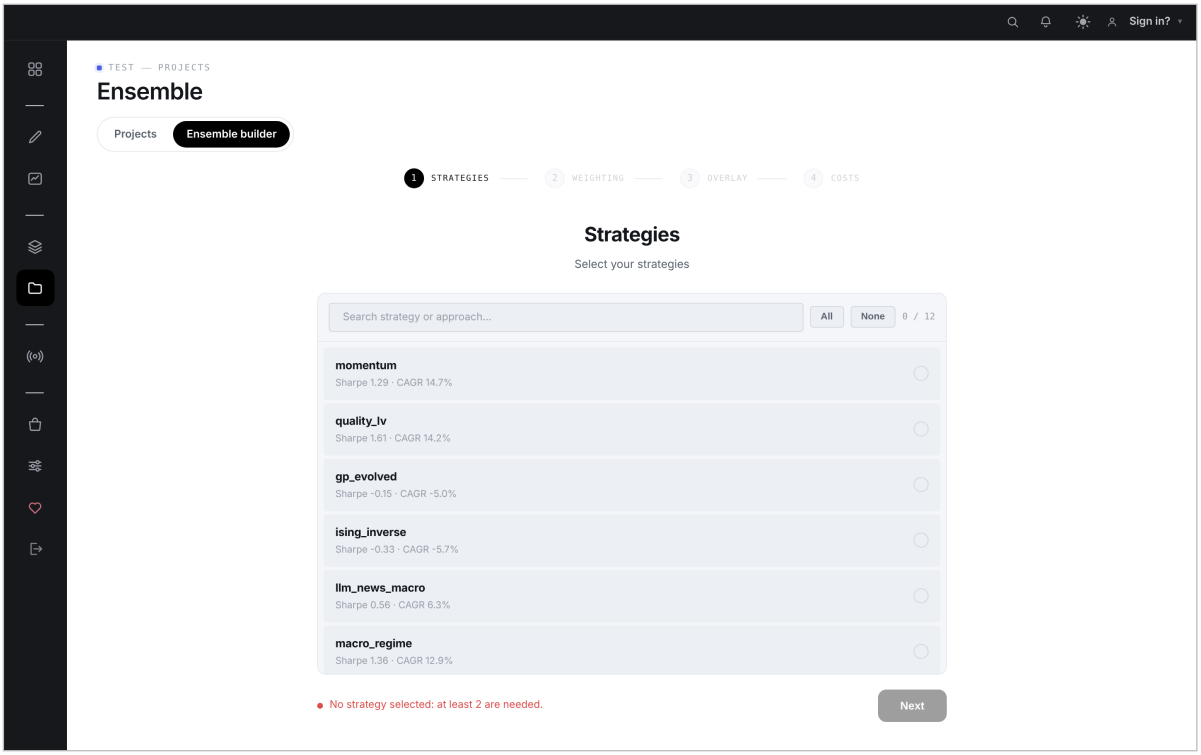


Figure 5.2. Choosing the streams. Next to each one are its Sharpe ratio and return, so you do not have to look them up elsewhere.

Do not pick the best three

The temptation is to take the three highest Sharpe ratios. It is almost always the mistake: if the three strategies make the same bet, putting them together reduces nothing — you are just making it three ways. The **Corr** column on the home screen is there for this: two strategies with low correlation, even with modest Sharpe ratios, do better together than three good strategies that are all alike.

5.1.2 Step 2 — Weighting

How to share the capital among the chosen streams. Fourteen methods, grouped by family.

Family	The idea
Naive	All the same. **Equal**, an equal share each. Always worth trying: it is the benchmark the others are measured against.
Risk / covariance	Weights by how much each one swings and how much they move together. Whatever swings less weighs more.
Robust	As above, but without trusting the estimated correlations too much — on little data they are unstable.
Tail-risk	Looks at extreme losses instead of the average swing.
Return-aware	Also takes into account how much each one is expected to earn, not only how much it risks.
Regime-adaptive	Changes the weights according to the state of the market.

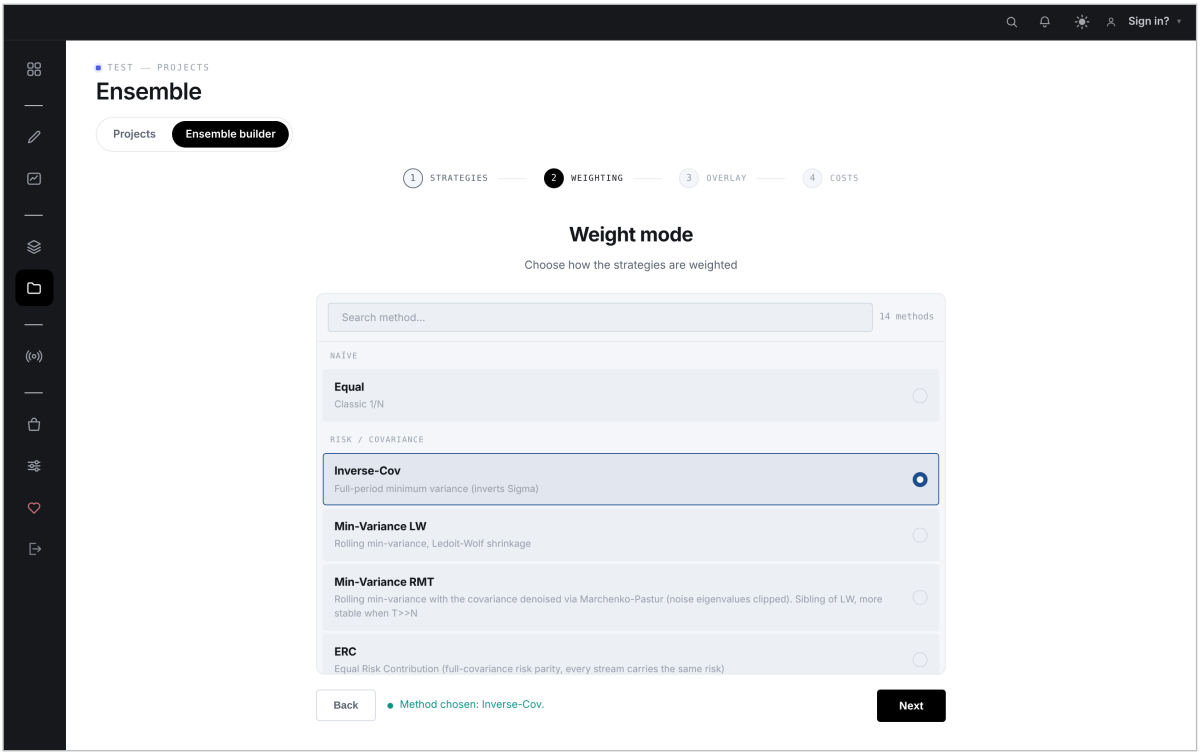


Figure 5.3. The weighting methods, by family. Each one carries its line of explanation.

Where to start

Equal. Then try **Min-Variance LW** or **ERC** and see whether the drawdown really falls. If a complicated method does not beat equal weighting, equal weighting is the answer: it has one parameter fewer to get wrong.

5.1.3 Step 3 — Overlay

Overlays reduce exposure when something happens. Several can be switched on together: their effects multiply, and the order in which you choose them does not change the result.

Overlay	Reduces exposure when...
Vol-Regime	realised volatility rises.
Jump-Vol	there is a sudden jump: it reacts within a day.
Corr-Gate	the strategies start moving all together, that is when diversification disappears.
Regime-GMM, **Iso-Stress**	market conditions look like those of past periods of stress.
Macro-Regime	the macroeconomic signals worsen.
News-Pressure	the volume of news goes above normal.

An overlay also takes away gains

A brake that reduces exposure reduces everything: the losses and the gains. It is worth looking at the two numbers together — how much the drawdown fell, and how much the return fell. Switching on five because 'it can't hurt' leads to a portfolio that is almost always out of the market.

5.1.4 Step 4 — Costs

The costs: commissions and slippage. The backtest subtracts them.

> **Raising the costs is the most honest check you can make**
>
> A strategy that stays good with the costs doubled is robust. One that stops working was earning inside the spread, and on a real account it will not earn.

5.2 Saving

At the end you give it a name and save it. From then on the project appears in the list, its performance is what the home screen draws, and — most importantly — it can be chosen on the **Live** page (Chapter 6).

6

Live

This is the page where a strategy stops being an exercise. A runner always starts in simulation: it works out the orders on the connected account and writes them to its journal, without sending them. Real orders are switched on runner by runner, when you decide, with an explicit confirmation. From then on, money moves.

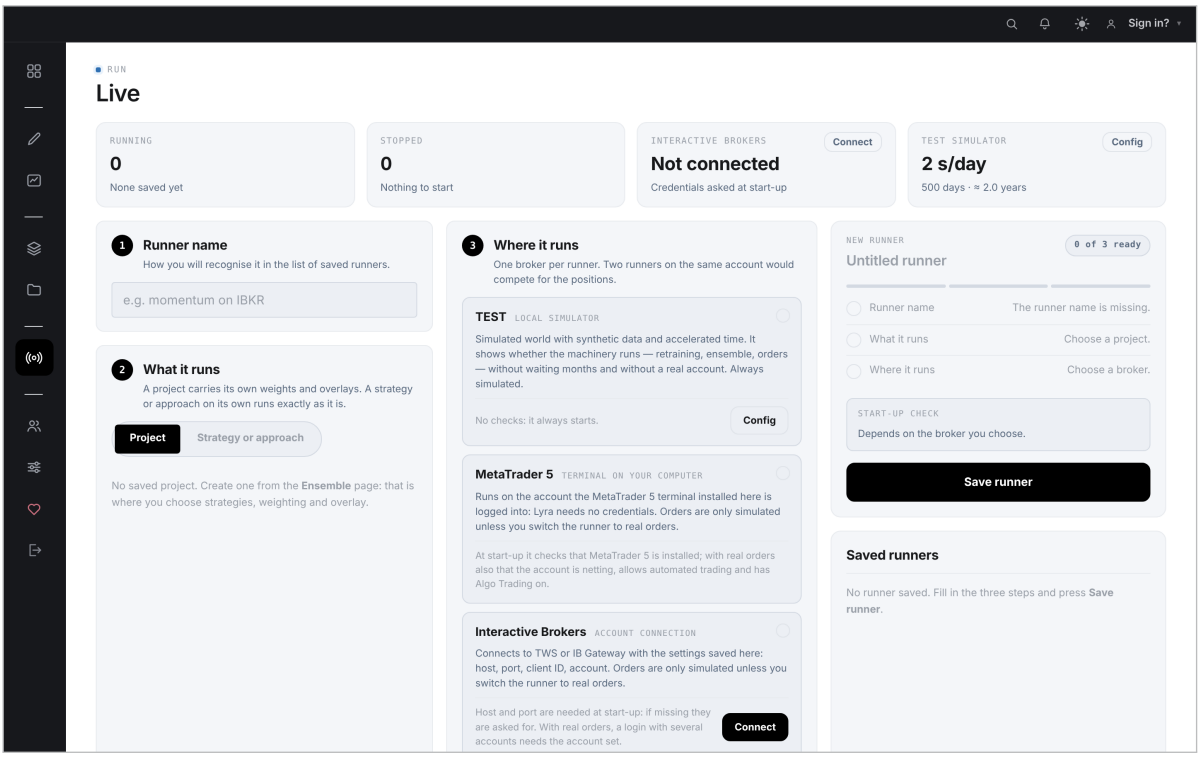


Figure 6.1. The Live page. At the top the status, below it the three steps to create a runner, on the right the summary of what is missing and the saved runners.

6.1 What a runner is

A **runner** is what runs a strategy on an account. It is made of three parts, and they are exactly what the three steps ask for:

1. **A name.** It only serves to recognise it in the list.
2. **What it runs.** A project (with its weights and its overlays), or a single strategy.
3. **Where it runs.** Which broker. For MetaTrader 5 and Interactive Brokers there is also the choice of orders: simulated or real (Section 6.3).

The panel on the right shows which of the three are in place: the **Save runner** button stays off until all three are. Below it, a line says what the orders will be like, already in the colour they

will have on the runner's card.

6.2 The three brokers

Broker	What it does
TEST	A simulated world with synthetic data and accelerated time. It needs no account and no credentials. It shows whether the machinery works — retraining, composition, orders — without waiting months. It is always simulated.
MetaTrader 5	Runs on the account the MT5 terminal installed on the computer is logged into: Lyra needs no credentials. At start-up it checks that MT5 is there; with real orders, also that the account is *netting*, that the broker allows automated trading and that **Algo Trading** is on.
Interactive Brokers	Connects to TWS or IB Gateway with the host, port, client ID and account saved with Connect. If they are missing, it asks for them at start-up.

The order to follow, without exceptions

1. **TEST** for a few simulated days. It shows that the machinery starts, that orders come out, that the journals get written.
2. **The real broker, in simulation.** The runner reads the account's equity and positions and writes to the journal the orders it would send. You find out which instruments the broker does not trade and which quantities do not add up.
3. **A demo or paper account, with real orders on.** Real orders, play money: you find out about trading hours, rejections and missing permissions.
4. **A real account**, with a fraction of the capital.

Skipping a step saves no time: it only means finding the same problems later, with the money in.

6.3 Simulation or real orders

Once MetaTrader 5 or Interactive Brokers is chosen, the **Orders** panel appears below the broker cards, with two positions: **Simulation**, the starting one, and **Real orders**.

Switching real orders on

1. In step 3, choose **MetaTrader 5** or **Interactive Brokers**.
2. In the **Orders** panel, press **Real orders**. A window opens that says what changes.
3. Read it, tick the box and press Enable real orders: the button stays off until the box is ticked.
4. Save runner. The **REAL ORDERS** label appears in red on the runner's card.

For a runner that is already saved, use **Switch to real orders...** on its card, with the runner stopped: the window is the same.

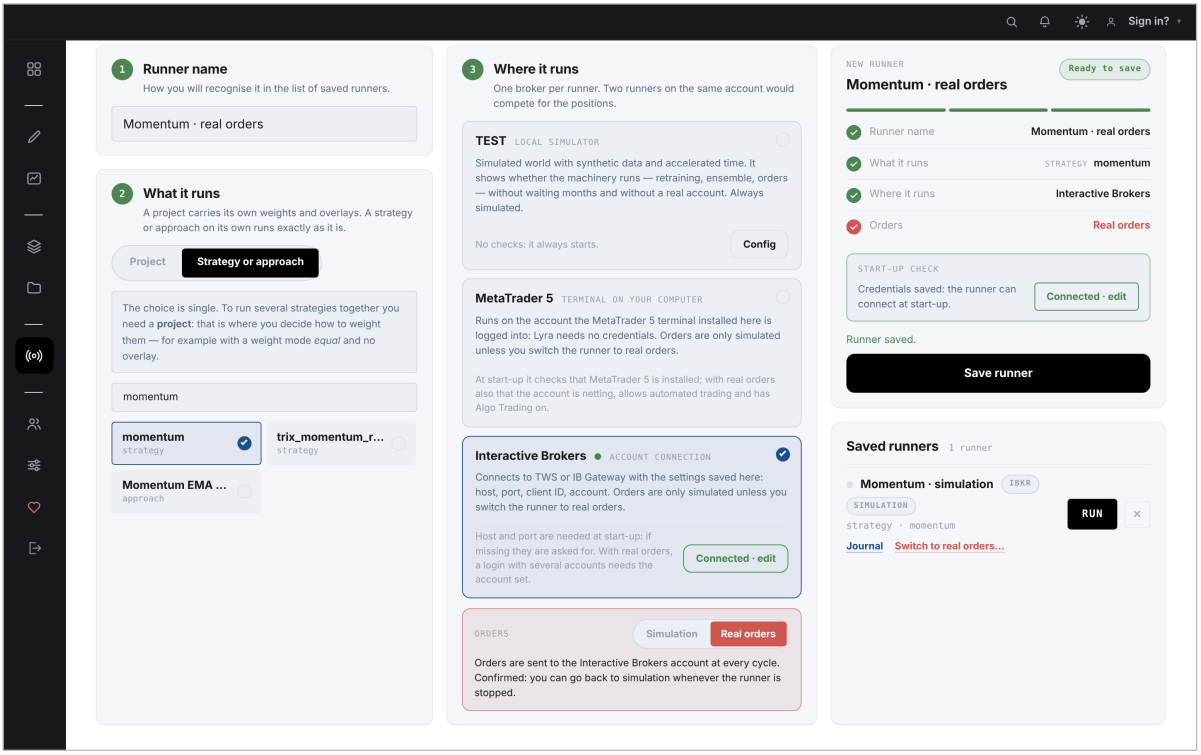


Figure 6.2. A runner with real orders on: the **Orders** panel and the summary line turn red. On the right, a runner already saved in simulation.

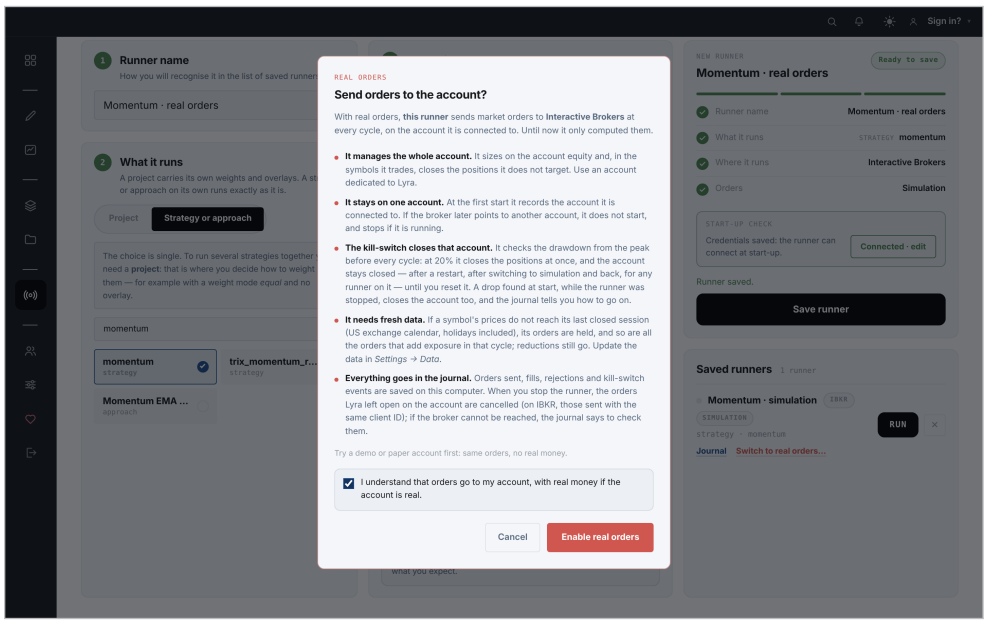


Figure 6.3. The confirmation of real orders. The button turns on only after the box is ticked.

The runner manages the whole account

It sizes the orders on the equity of the whole account, not on a part of it, and in the symbols it trades it closes the positions that are not among its targets: a share bought by hand on that account, if it is in the strategy's universe, can be sold at the next cycle. That is why the account must be dedicated to Lyra — one per runner, no manual trading.

First reduce, then increase

In every cycle the runner first sends the orders that reduce positions, then those that increase them; a reversal (from short to long, or the other way round) becomes two orders, the close and the open. If part of the book cannot go out — old data on a stock, an order still open at the broker, a holding with no price — or a reduction does not go through, the orders that increase exposure wait for the next cycle. A cycle left halfway never leaves the account more exposed than before.

The mode is always visible: on the runner's card (**SIMULATION** or **REAL ORDERS**) and on every line of the journal. It can only be changed with the runner stopped, and going back to simulation with **Back to simulation** asks for no confirmation. The **RUN** button of a runner with real orders is red and asks once more before starting.

The confirmation stays on record

The runner records when real orders were confirmed, and the journal reports it. Going back to simulation deletes the confirmation: to switch them on again, you confirm again.

6.3.1 The confirmation is for one account

At the first start after the confirmation, the runner records the account it is connected to — on MT5 the number and server, on IBKR the number — and from then on it does not move. If later the MT5 terminal is logged into another account, or the **Connect** settings point to another IBKR account, the runner does not start and the journal says which account it expected and which one it found. If it happens while it is running, it stops before sending anything else. The account is shown on the runner's card and in the start window.

Moving to another account

1. With the runner stopped, **Back to simulation**.
2. Connect the broker to the new account (the MT5 terminal, or **Connect** for IBKR).
3. **Switch to real orders...** and confirm: at the next start the runner binds itself to the new account.

Accounts in another currency

Lyra sizes the orders in dollars, on the prices of its own data. An account in euros is fine: the equity is converted with the exchange rate the broker gives (on MT5, the EURUSD price in the terminal). If there is no rate, real orders do not start and the journal says so.

6.4 The journal

Every runner keeps a journal, which opens with **Journal** on its card. Everything the runner does goes into it, one line per event, with the time and the mode in which it happened:

- starts and stops, with the account it connected to: number, type (demo, paper or real), terminal; with real orders, also the check that it is the confirmed account and Lyra's orders left open, cancelled at start and at stop;
- the orders. In simulation, those it would send; if they are the same at the next cycle, a single line says so. With real orders, those sent and their outcome: filled with quantity and price, partly filled, still working (*pending*), cancelled, rejected with the reason the broker gives;
- warnings, for example a symbol the broker does not trade, or data that are too old;
- the kill-switch events (Section 6.5);
- maintenance: retraining and data updates.

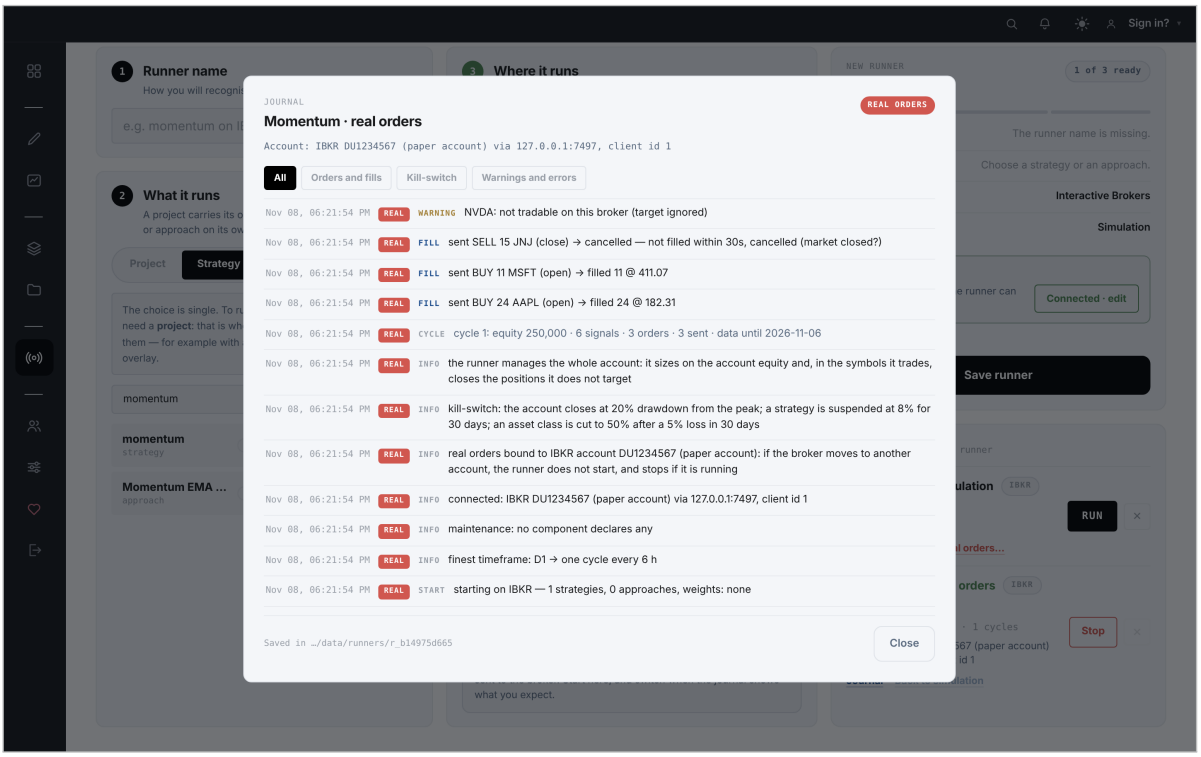


Figure 6.4. The journal of a runner with real orders: two orders filled, one cancelled because it was not filled in time, a warning.

The filters at the top separate orders and fills, kill-switch, warnings and errors. While the runner is running, the journal updates by itself.

> **It lives on the computer**
>
> The journal is a file in the application folder, `data/runners/<id>/journal.jsonl`, one line per event; beyond 5 MB it continues in `journal.1.jsonl` and so on, up to ten files. It stays after the application is closed and even after the runner is deleted: it is the account's logbook, and it is worth keeping.

6.5 The kill-switch

Three levels, on by default, the same as in the backtest:

Level	When it trips	What it does
Strategy	An 8% drawdown	Suspends it for 30 days, then it comes back by itself.
Asset class	A 5% loss in 30 days	Halves its exposure for 20 days.
Account	A 20% drawdown from the peak	Closes the positions in the runner's symbols and opens no others.

A suspended strategy and a halved asset class stay that way even after the runner is restarted, until they expire.

The account level, with real orders, belongs to the **account** and not to the runner. The drawdown is measured at the start of every cycle, before deciding: if the fall happened between one cycle and the next, the runner does not send another round of purchases, but closes the positions in the symbols it trades straight away. If something stays open (market closed, a rejection) it tries again at once, then every minute three times, then at the normal pace; once the positions are closed, it stops by itself.

From then on the account stays closed: after a restart, after going to simulation and back to real orders, for any runner with real orders on that account. Starting a runner on a closed account only serves to close the positions left: the start window says so, the runner closes and stops. While it is closing, not even repeated errors stop it.

Reopening the account

1. Stop the runner.
2. **Journal** → (Reset kill-switch), and confirm. The account reopens for every runner that uses it.
3. At the next start the peak starts again from the account's equity.

Why it belongs to the account, and is saved

The kill-switch protects money, and the money is in the account. When it belonged to the runner, a restart was enough to start again from the equity of the moment — after a 15% drawdown the 20% limit in fact became 32% — and going to simulation and back, or using another runner, was enough to reopen a closed account. The state lives in `data/accounts/`, one file per account; if it cannot be read, real orders do not start: a closed account must not reopen because of a damaged file.

A fall found at start-up, withdrawals and deposits

The kill-switch looks at the account's equity and does not tell a loss from a withdrawal. If at start-up the equity is already below the account's peak by more than the threshold — the fall happened with the runner stopped — the kill-switch trips and is recorded, but the runner sends no orders and says how to go on. If it is a loss, start the runner again: it closes the positions left and stops. If money was withdrawn, reset the kill-switch from the journal instead. If the account's equity cannot be read at start-up, real orders do not start.

In simulation

The kill-switch trips in simulation too, and the journal says what it would have done. There it closes nothing and does not touch the account: it belongs to the runner, and it is reset the same way.

6.6 When orders do not go out

When an order does not go out, or does not go as expected, the journal says so in one line. The most common ones, what they mean and what to do:

In the journal	What it means, and what to do
'orders NOT sent: the data are too old'	The prices the runner decides on are those downloaded in Settings → Data, and they do not update by themselves. Every symbol must reach the last closed session of its market: for US shares and ETFs, the New York Stock Exchange calendar, holidays included; for forex and crypto, one session is tolerated. A day's bar is expected from the following day. The orders on the symbols that are behind wait, and so do those that increase exposure; reductions on the other symbols go out. Update the data: the held orders go out at the first cycle with fresh data. The closes ordered by the account's kill-switch go out regardless.
'real orders cannot start on this account'	The account cannot receive orders from Lyra, and the line says why: on MT5, a *hedging* account (a *netting* account is needed), trading disabled or not allowed to expert advisors, **Algo Trading** off, the Python API switched off in the terminal's options; on IBKR, a login with several accounts and no account given in Connect; on both, an account in another currency with no rate to convert it.
'the real orders of this runner were confirmed for . . . '	The broker is connected to an account different from the one the runner started on after the confirmation. If it is the right account, go back to simulation and switch real orders on again (Section 6.3.1).
'cycle skipped, nothing sent'	An obstacle that appeared while the runner was on, for example **Algo Trading** switched off or the account data not yet in sync. The runner skips the cycles without sending anything and resumes by itself when it goes away.
'kill-switch: this account was closed'	The account was closed by the kill-switch: the runner starts only to close the positions left, then stops (Section 6.5).
'an order is still working at the broker'	On that symbol there is an order still open — Lyra's or one placed by hand. While it stays open, the runner does not send another: when the first was filled, the position would double.

In the journal	What it means, and what to do
'order(s) that add exposure not sent'	Part of the book did not go out (one of the lines above, or a holding with no price in that cycle) or a sale did not go through: in that cycle only the reductions go out, and the purchases are worked out again at the next cycle.
An order *rejected*	The reason is the broker's, written in the line: permissions, market closed, quantity below the minimum. In TWS and IB Gateway the **Read-Only API** option must be off, otherwise every order is rejected.
An order *cancelled* on IBKR	It was not filled within 30 seconds, usually because the market was closed, and the runner cancelled it: left open, it would have been filled later while the runner was working out another one.
An order *pending*	The broker accepted it but has not filled it yet (MT5), or its cancellation is not yet confirmed (IBKR): it can still be filled. The runner sends no other orders on that symbol while it stays open, and cancels it when it stops.
An order *skipped*	The quantity is below the broker's minimum lot, and nothing went out: the difference is smaller than one lot.
'the broker is not answering'	Connection dropped, TWS or MetaTrader 5 closed or restarting. The runner sends nothing and keeps trying without giving up, with waits that grow up to a quarter of an hour; when the broker answers again, it says so.
'cycle failed (1 of 3 in a row)'	An error of another kind. The runner tries again within a minute; three errors in a row stop it, with the cause in the journal — except when it is closing an account closed by the kill-switch.
'cannot connect to TWS/IB Gateway'	TWS or IB Gateway off or not logged in, API not enabled (Configure → API → Settings), wrong port, or another program is using the same client ID.

6.7 The simulator

The **TEST SIMULATOR** panel at the top says how long a simulated day lasts — in the screenshot, two seconds — and how much time it will cover in all. Pressing Config changes the pace.

> **What speeding up time is for**
>
> A strategy that retrains every thirty days, on a real account, retrains for the first time in thirty days. In the simulator it happens in a minute, and if the retraining has a problem you find out at once instead of next month.

6.8 Credentials

The broker credentials stay **on the computer of whoever uses the application**, in a local file. They do not go through the site, they do not end up in any database and they do not leave the machine.

For Interactive Brokers, Connect saves the host, the port, the client ID and the account. Port and client ID must be numbers, and the window says so at once. The usual ports are 7 497 for TWS in paper and 7 496 for real, 4 002 and 4 001 for IB Gateway. The account must be given if the login manages more than one.

> **Two consequences**
>
> The first: nobody can recover them for you: if they are lost, you set them up again with the broker. The second, more important: whoever has access to that computer has access to the account. The application is meant to run on a personal machine, not on a server others can reach.

6.9 Starting and stopping

Once saved, the runner appears in the list at the bottom right, with the commands to start and stop it. The two counters at the top — **RUNNING** and **STOPPED** — say how many are in each state.

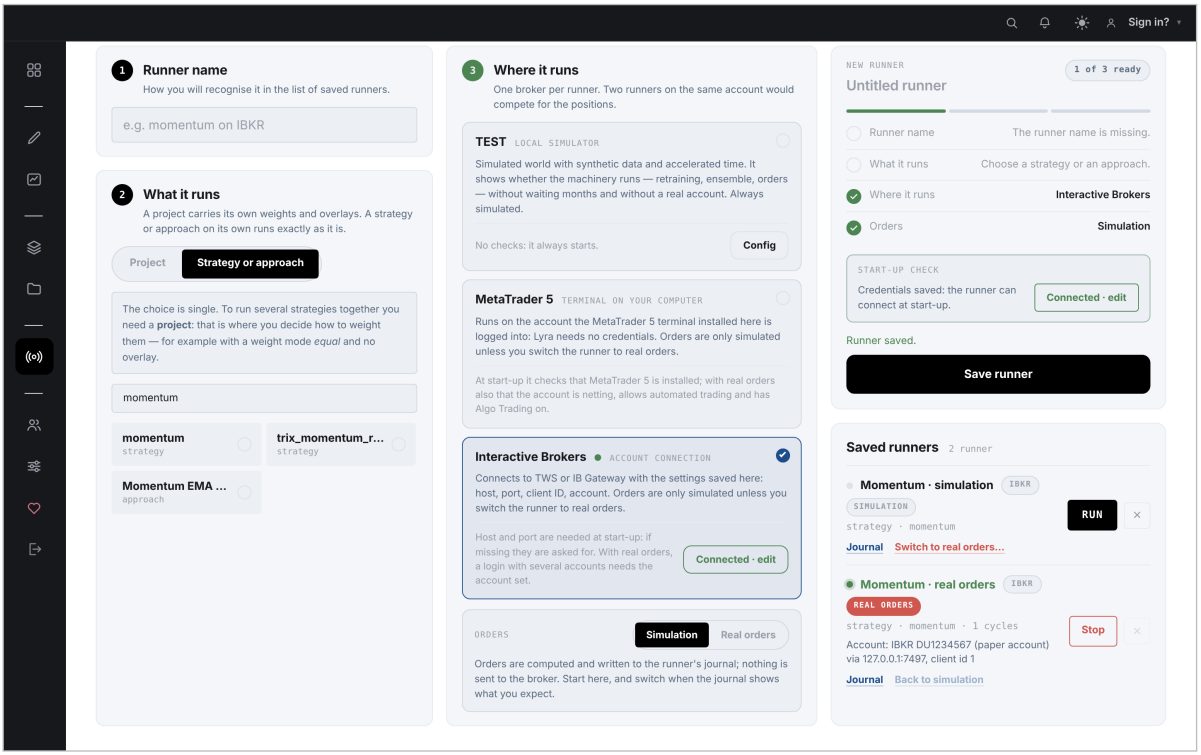


Figure 6.5. Two runners on the same model: one stopped in simulation, one running with real orders.

The first cycle starts at once. The following ones follow the shortest bar among those the runner watches, a quarter of a bar at a time: for daily strategies, one cycle every six hours. Asking the broker more often would produce identical cycles.

With real orders, RUN asks for confirmation once more and says which account the orders will go to; if the account was closed by the kill-switch, it says that starting will only serve to close

positions.

Stopping: no new orders

Once **Stop** is pressed, the orders of the current cycle not yet sent do not go out any more; those already sent reach their outcome, which goes into the journal. With real orders, Lyra's orders still open at the broker are cancelled: none of them is filled later without the runner watching.

One broker per runner

Two runners on the same account fight over the positions: the first opens, the second sees a position it does not expect and closes it. The page says so in step 3, and does not let a second runner start on the same broker while the first is running.

7
Community

It is the area where users meet, and the only one in the application that needs an account. In the menu it sits under **Share**; the page has four tabs:

Tab	What it is for
Marketplace	Buying and selling strategies, approaches, overlays and weighting methods. It is the subject of this chapter.
Messages	Conversations with other users, for example with the author of a strategy.
Feedback	The reports and suggestions sent to the Lyra team.
My account	Profile, security and subscription (Chapter 9).

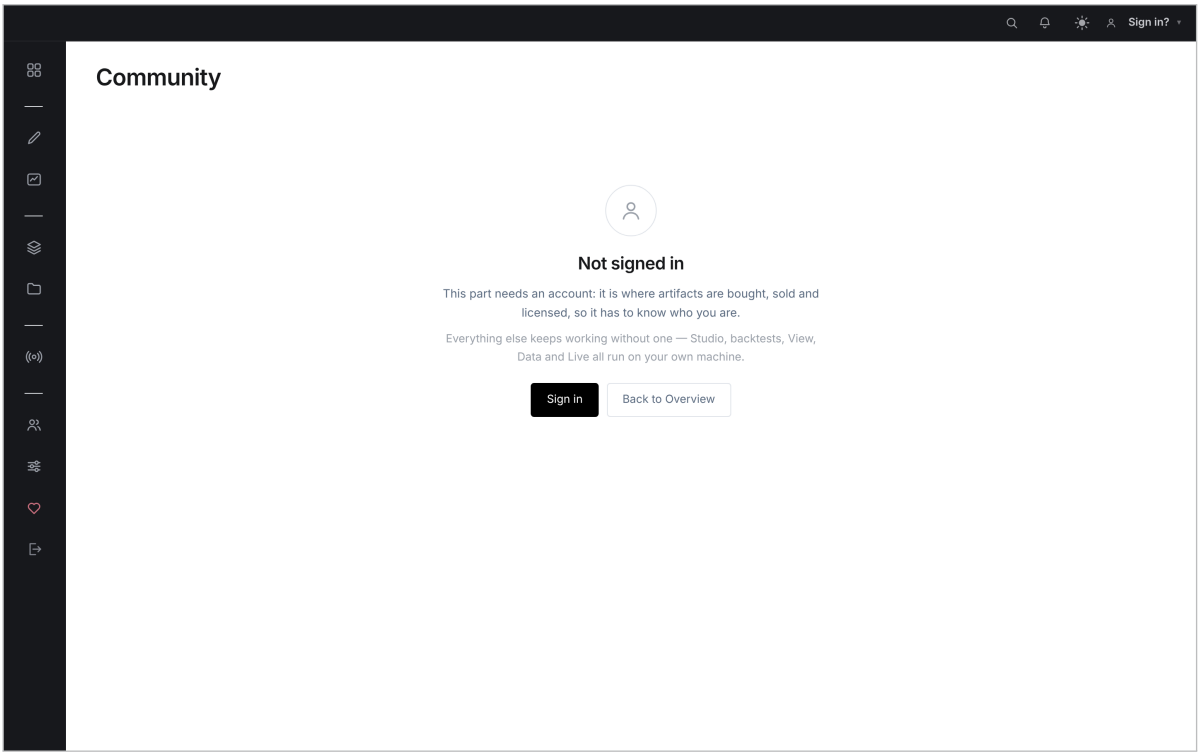


Figure 7.1. The **Community** without an account. The rest of the application keeps working: the page says so.

7.1 Why an account is needed here and not elsewhere

Because here there is someone on the other side. Building and testing strategies happens entirely on your own computer, and asking for an account to do it would make no sense. Buying, selling

and granting a licence require the system to know who is who.

7.2 Buying

From the storefront to the system

1. Open the artifact's page: description, metrics declared by the seller, comments, sometimes a video.
2. **Buy** — if it is free, access is immediate; if it is paid, the payment flow opens.
3. Once the payment is done you go back to the application. If access does not show as active yet, **it is not an error**: the confirmation arrives a moment later. The page says so and asks you to wait.
4. **Install** — the artifact is written to the system, and from then on it appears in the lists like all the others.

Before installing, the conflicts

If something with the same name already exists in the system, the application says so *before* writing. It is worth reading: an installation that overwrites a strategy you were working on is lost work.

7.2.1 If a purchase does not show up

There is a command for exactly this, on the artifact's page, that explains why it does not show as purchased. It tells apart two things that at first sight look the same: 'you did not buy it' and 'I could not ask'. They are opposites, and the second is solved by trying again.

If the payment went through but access does not open, there is a reconciliation command: it checks the payment again with the provider and, only if it shows as collected, completes the operation.

7.3 Selling

To publish you need two things, in this order:

1. **The marketplace must be open.** During the beta it is closed: nobody uploads anything. Items already in the storefront stay visible.
2. **A paid plan.** With the marketplace open, any paid plan will do, from Pro up (Chapter 9). On Free you can browse, download and buy, but not sell.

Why a subscription even to publish for free

It is not about the money. Whoever publishes puts their name next to a strategy that others will run with their own money: a filter at the entrance is worth more than any check after the fact. Someone without a subscription remains a full user — they browse, download, buy and use. They just cannot put things in the storefront.

A published artifact goes online at once: there is no approval queue. The description and the video can be corrected later; the bundle of files cannot.

7.4 If an artifact does not perform as declared

You open a dispute from the artifact's page. It is not a discussion: the system runs two automatic checks.

Check	What it compares
Integrity	The installed files with those sold, one by one.
Performance	The metrics declared by the seller with those obtained by running the calculation again on your own system.

The case is a conversation between buyer, seller and the administration, with the results of the two checks attached. It can be withdrawn at any time.

> **Two different results do not mean a deception**
>
> A strategy can perform differently because the data are different: a different universe of instruments, a different period, different costs. The check is there to establish *by how much* they differ, which is the starting point of any sensible discussion.

8

Settings and data

The **Settings** item gathers three tabs: **Config**, **Data** and **Market rules**. They are the three places where you tell the system what to work with, and within which limits.

8.1 Data: bringing in the prices

Without data, every page of the application is empty. It is the first thing to do after installing.

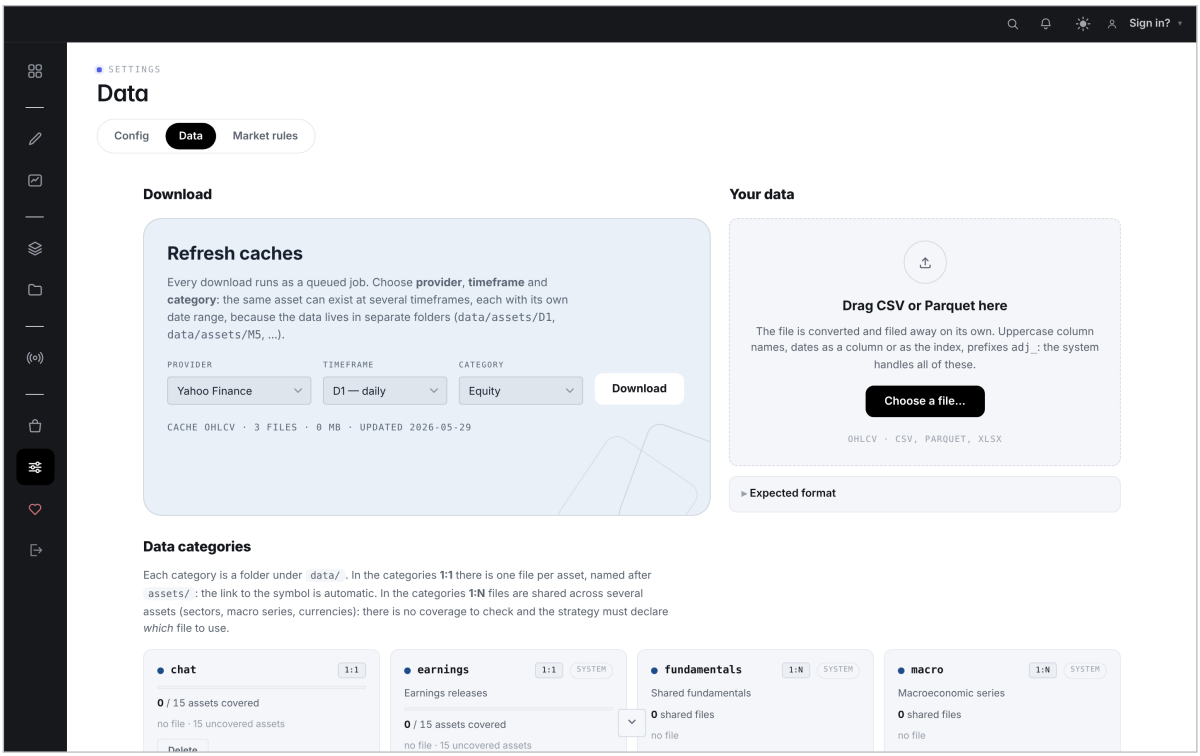


Figure 8.1. The Data page. On the left the automatic download, on the right the manual upload, below them the categories.

8.1.1 Downloading

You choose three things and press **Download**:

Provider
: Where to get them from. Some need a personal key, which you set up from the site.

Timeframe
: The time step. The same instrument can exist at several steps: daily data and five-minute data live in different folders and do not get in each other's way.

Category
What kind of data it is.

The job goes into the queue and the bell tells you when it is done. Below the fields, a line says what is already there: how many files, how much space, up to when they go.

8.1.2 Uploading your own file

The **Your data** panel accepts CSV, Parquet and XLSX files dragged onto it. The system recognises by itself columns written in capitals, dates given as an index or as a column, and prefixes such as *adj_*. The **Expected format** panel shows what it expects.

8.1.3 The categories

Each category is a folder, and they come in two types.

Type	It means
1:1	One file per instrument, named after the instrument. The link is automatic, and the application can say how many instruments are covered and how many are not.
1:N	Files shared by several instruments: sectors, macroeconomic series, currencies. Here there is no coverage to compute, and the strategy has to declare which file to use.

> **'0 / 15 assets covered' is not a fault**
>
> It means the category exists but has no data yet for any instrument. It fills up by downloading, or by dragging files into it.

8.2 Config: the limits

These are the parameters that apply to every calculation.

Field	What it limits
Portfolio & allocator	
Initial capital	The capital the calculations start from.
Annual target volatility	How much you accept the portfolio swinging in a year. It is the main risk dial.
Max weight per position	The most a single instrument can weigh.
Max gross per asset class	The most a whole class can weigh — shares, bonds, currencies. It guards against diversification that is only apparent: four bonds all going the same way are a single position.
Max gross leverage	The total exposure. 1.0 means no leverage.
Ensemble	
Target volatility	The volatility every stream is brought to *before* being weighted.
Regime window, **thresholds**	How the state of the market is assessed for the modulation.

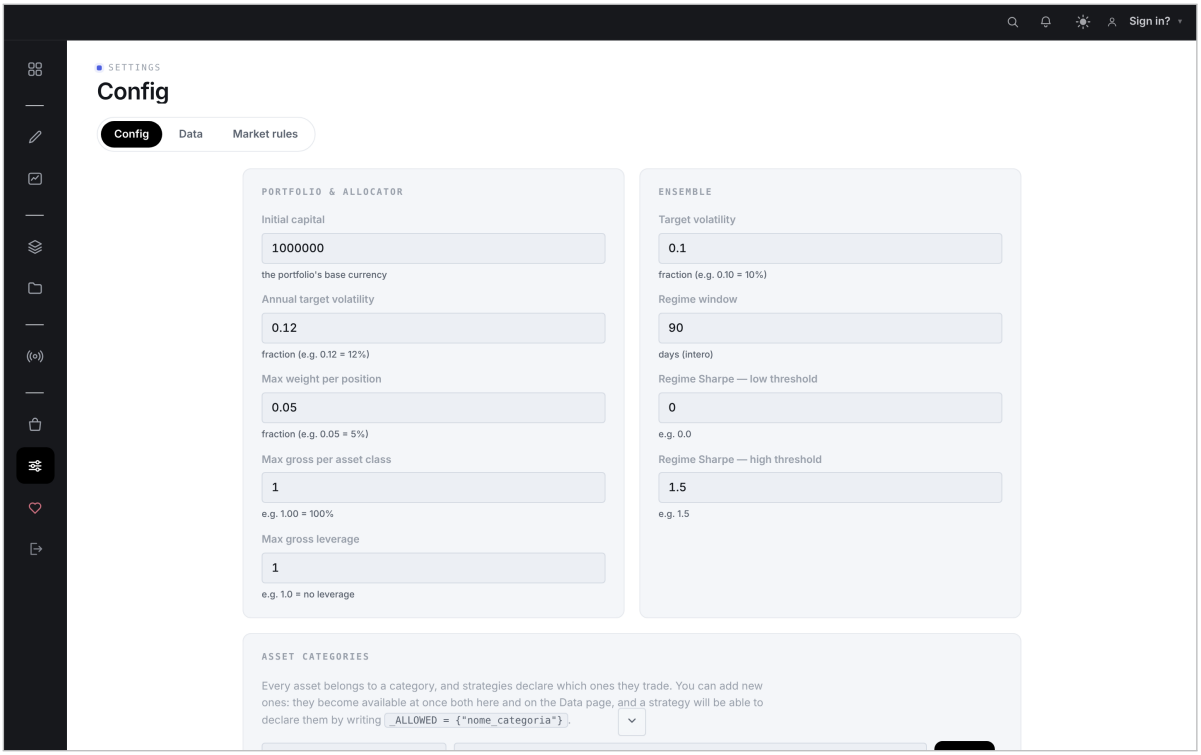


Figure 8.2. The parameters. Each field carries its unit below it, because the difference between 0.12 and 12 is a factor of a hundred.

> **Target volatility also acts on the real account**
>
> It is not a display setting. The same factor that normalises the streams in the backtest multiplies the positions the runner opens on a real account. Changing it changes how much you really risk, not just what you see on the chart.

Further down, **Asset categories** lets you add categories of instruments: once created, a strategy can declare them and restrict itself to them.

8.3 Market rules: the regimes

A **regime** is the state of the market, with three values: **bull**, **neutral**, **bear**. Strategies can declare that they weigh things differently according to the state.

8.3.1 How to read the page

Each panel at the top is a *market* — a set of instruments that share the same way of measuring the regime:

RULE
: The rule used to compute the state.

DRIVER
: The series that drives the decision — for shares, SPY; for forex, the dollar.

SYMBOLS
: How many name patterns fall into this market.

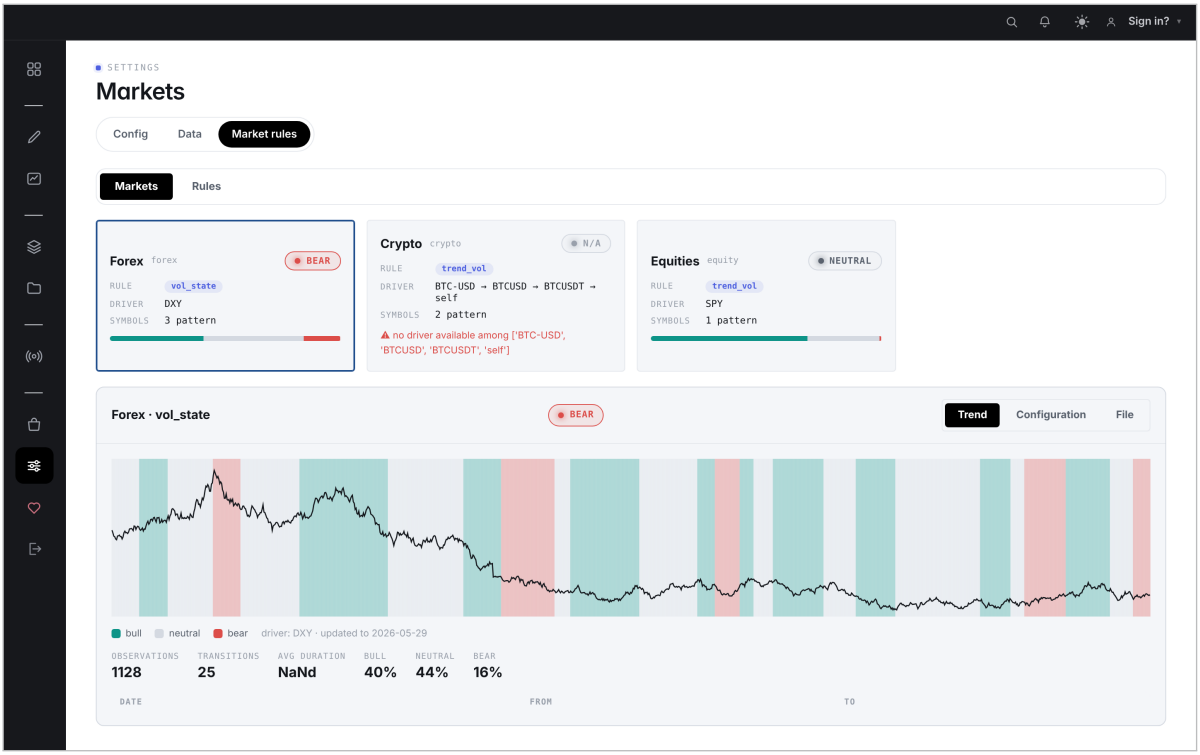


Figure 8.3. The configured markets. Below them, the history of the states: green bull, grey neutral, red bear.

The coloured bar

How much time was spent in each state.

'No driver available'

The panel of a market that cannot find its driving series says so with a warning, like the **crypto** market in the screenshot. That market **modulates nothing**: it is not an error that blocks anything, but the strategies on those instruments behave as if the regime were always neutral. The fix is to download the missing series from the **Data** page.

8.3.2 The chart

It shows the driving series with the background coloured by state. Below it, the counts: how many observations, how many times the state changed, and the percentage of time spent in each.

Few changes of state is a good thing

A regime must be a *state*, not a reaction. Twenty-five transitions in five years mean the rule tells different periods apart; two hundred would mean it is chasing noise, and a strategy that changes weight every week pays costs without taking a position.

9

Account and subscription

9.1 What the account is for

Four things, and nothing else: entering the marketplace and receiving messages — both in the **Community** — having a public profile, and taking out a subscription. The rest of the application works without one.

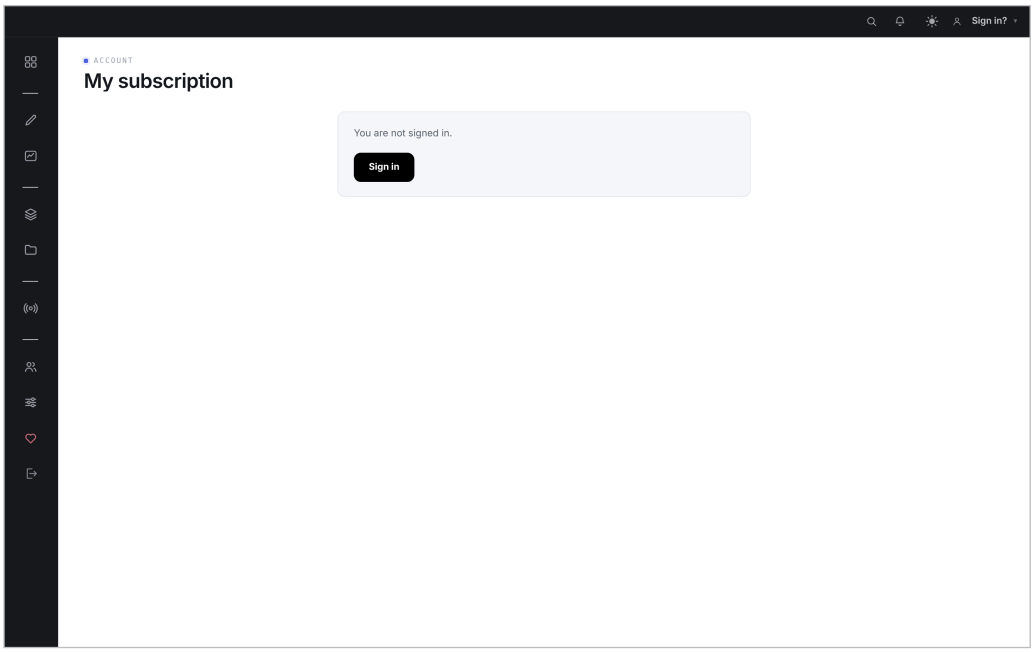


Figure 9.1. Pages that need an account say so and offer to sign you in, instead of disappearing from the menu.

9.2 Signing in for the first time

During the beta

During the beta, access is reserved for people who signed up on the site. The first sign-in is not a registration: it is a password reset.

1. On the sign-in page, choose **Forgotten your password?**.
2. Enter the address you used to sign up for the beta.
3. A message arrives with a code; use it to set a new password.
4. From then on you sign in normally.

An address that is not among the sign-ups is refused, both at sign-in and at registration.

You can also sign in with Google (**Continue with Google**): the sign-in opens in the system

browser and the application picks up the result.

9.3 The profile

Profile, in the account menu at the top right, gathers the account on one page. **Sign out** is at the top right of the page.

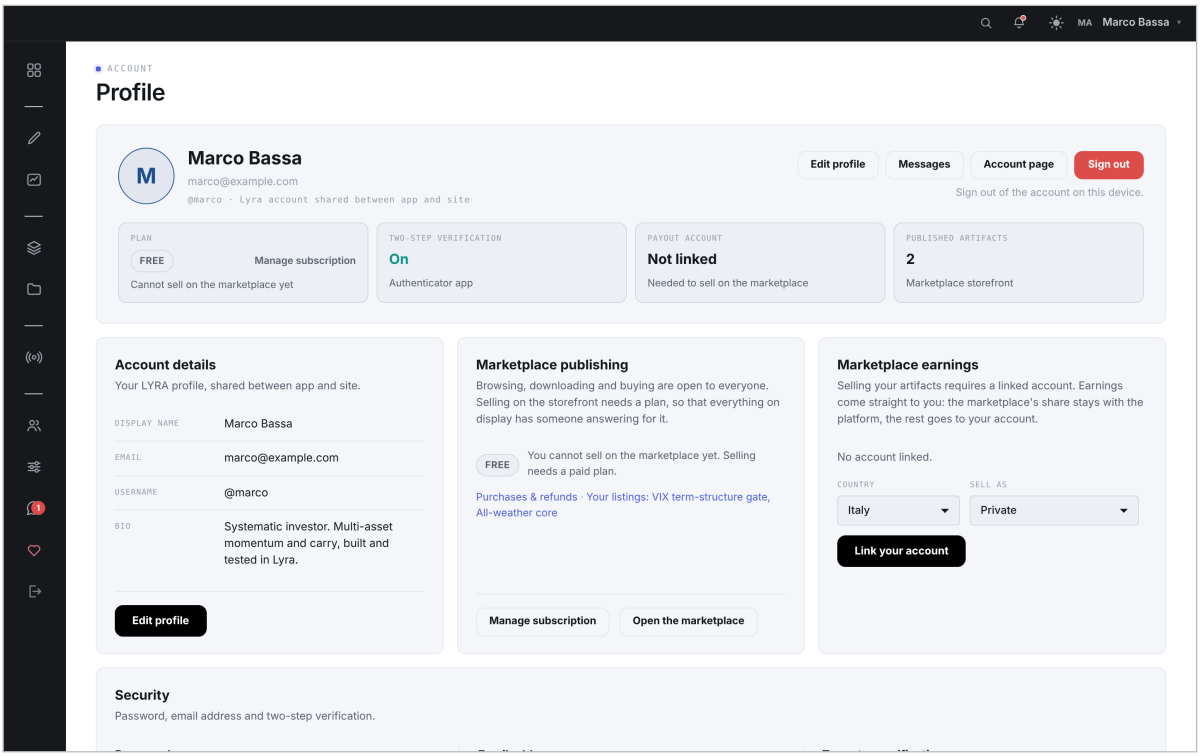


Figure 9.2. The top of the profile: the plan, two-step verification, the payout account and the published artifacts at a glance; below, the account details, marketplace publishing and earnings.

Account details
: Display name and bio — what others see next to a published artifact — changed with Edit profile.

Marketplace publishing
: Whether you can sell, with links to your purchases and your listings, and to **My subscription**.

Marketplace earnings
: The payout account that receives what you sell. Selling for a price needs one, and linking it needs two-step verification first.

Security
: The password, the email address and two-step verification. They are not stored by the application: they live on the site, and the application only passes them on.

Preferences, Brokers and data keys
: The theme and the notifications; the broker connections and the data keys, which stay on this computer.

Send us feedback
: A report or a suggestion for the Lyra team.

9.4 The plans

The plans are the same as on the Pricing page of the site. Free is the whole application; you pay for privacy, data and a higher share of what your strategies earn — not for using the application. They are on the **My subscription** page, in the account menu at the top right.

Plan	Price	What it adds
Free	€0, forever	The whole application: building and backtesting locally with your data, the local AI copilot, out-of-sample validation, live runners and kill-switch. On the marketplace: browse, download, buy. Share of earnings 0%: an allocated strategy enters the track record.
Pro	€49 a month or €490 a year	Full privacy for code and metrics; selling on the marketplace and buying at −10%; the institutional-data sandbox (from 2030); 10% of what your allocated strategies earn.
Professional	€149 a month or €1,490 a year	Raw history on your machine (from 2030) with −10% on data vendors; marketplace −20%; sandbox on more markets; 30% of earnings. It is the recommended plan.
Quant	from €200 a month or €2,000 a year, plus data	Exchange-licensed real-time (from 2032), in per-market packages; extended sandbox; 30% of earnings.
Desk	€8,000 a year	Five seats, installation on your servers, compliance export, white-label and SLA; 30% of earnings, per seat, by contract. Sold by contract, after a demo.

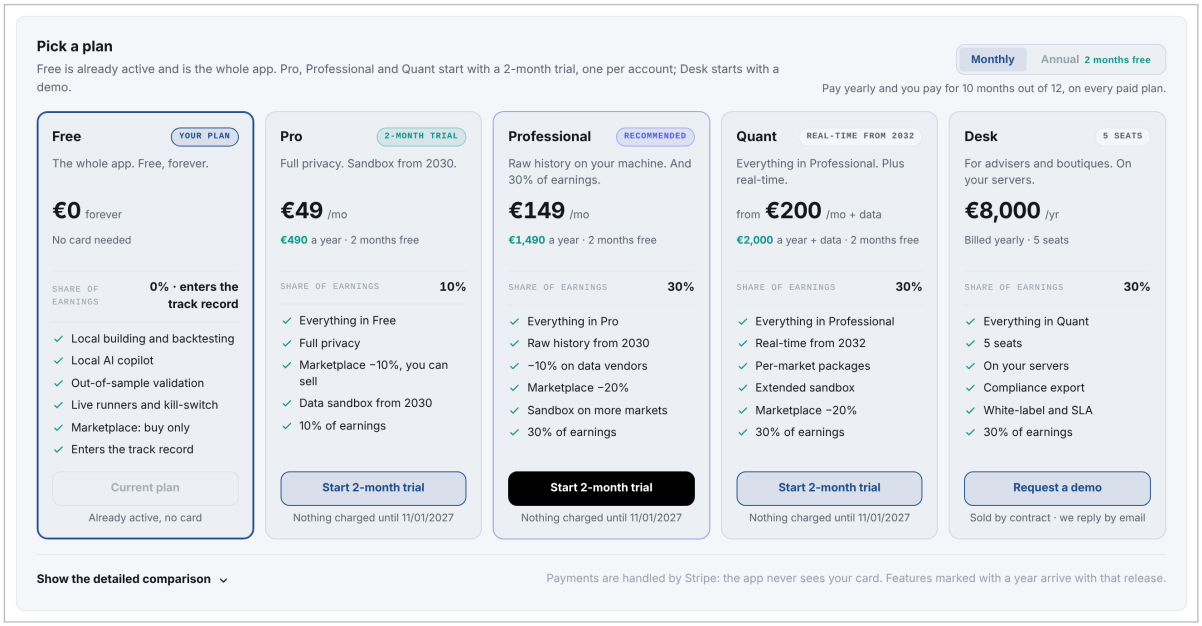


Figure 9.3. The plans on **My subscription**: price, share of earnings and what each plan adds. **Show the detailed comparison** opens the table with every feature, plan by plan.

The features with a year

Some features arrive with a later release: the sandbox and the raw history in 2030, real-time in 2032. The cards and the comparison table show the year next to them.

During the beta

The paid plans open after the beta. Until then Free is the whole application, nothing can be bought or sold on the marketplace, nothing is charged, and the plan buttons say **On sale after the beta**. A Desk demo can be requested already.

The share of earnings

It is the part of what a strategy earns, once it is allocated to capital, that goes to its author: 10% with Pro, 30% with Professional, Quant and Desk, set by contract. On Free an allocated strategy enters the track record without a share. Every submission is opt-in, strategy by strategy and under a contract: nothing is submitted without your consent. The site describes the steps, from the protocol to the live track record.

9.4.1 The two-month trial

Pro, Professional and Quant start with a two-month trial, one per account.

Starting the trial

1. On the plan's card, press **Start 2-month trial**; the hint below the button says until when nothing is charged.
2. The window shows **Charged today** at €0, the day the trial ends and the price that applies afterwards. Choose monthly or yearly there, then press **Start the trial**.
3. The Stripe page opens: Stripe asks for your card but charges nothing. The page updates by itself and shows **Trial until** and the date.
4. On that date the first payment is taken. Cancel before, and nothing is ever charged.

Changing plan during the trial costs nothing, and the trial keeps its end date. An account that has already had a trial, or a paid plan, pays from the first day: the button then says **Choose** followed by the plan's name.

9.4.2 Desk

Desk is not bought with a card: it starts with a demo on your firm's own strategies, and a contract. **Request a demo** on the Desk card opens a short form — the firm, how many people would use the platform, a message — and the request reaches the Lyra team, who reply by email and in **Messages**.

9.4.3 Monthly or yearly

Both. Yearly costs ten monthly payments: two months are free. Desk is yearly only. You choose when you subscribe and switch whenever you like, from the button next to your plan: Stripe works out the adjustment for the period already paid.

9.4.4 Changing plan

Upgrade and Downgrade on the cards change the plan at once: Stripe credits what you have already paid for the current period and charges the difference to the card on file. Before you confirm, the window lists what you gain or what you lose.

9.4.5 Cancelling, card, invoices

From Manage in Stripe on **My subscription**. The billing portal hosted by Stripe opens: there you cancel the subscription, change the payment method and download the invoices. The card never reaches the application.

> **Cancelling takes nothing you already have**
>
> The plan stays active until the end of the period you paid for. A purchased artifact stays purchased and installed, and strategies, code and data stay on your computer: you go back to Free, which is still the complete application. What stops is selling on the marketplace and the benefits of the plan.

9.5 When something does not add up

Symptom	What to do
'You cannot sell' but you have a subscription	Close and reopen the application: the link with the site is made at start-up.
The marketplace is empty	Same reason. If it persists, sign out and sign back in.
I paid and I cannot see the artifact	Wait a few seconds and reload. If it stays like that, use the command that explains why it does not show as purchased, on the artifact's page (Chapter 7).
The trial is not offered	The trial is one per account: an account that has already had one, or a paid plan, pays from the first day.
A page says it does not know, instead of showing a number	Not a fault: the application could not ask the site, and says so instead of making up an answer. Try again later.